Pharmacology of Antiepileptics

Antiepileptics (AEDs)
Antiseizure medications (ASMs)
Antiseizure drugs (ASDs)

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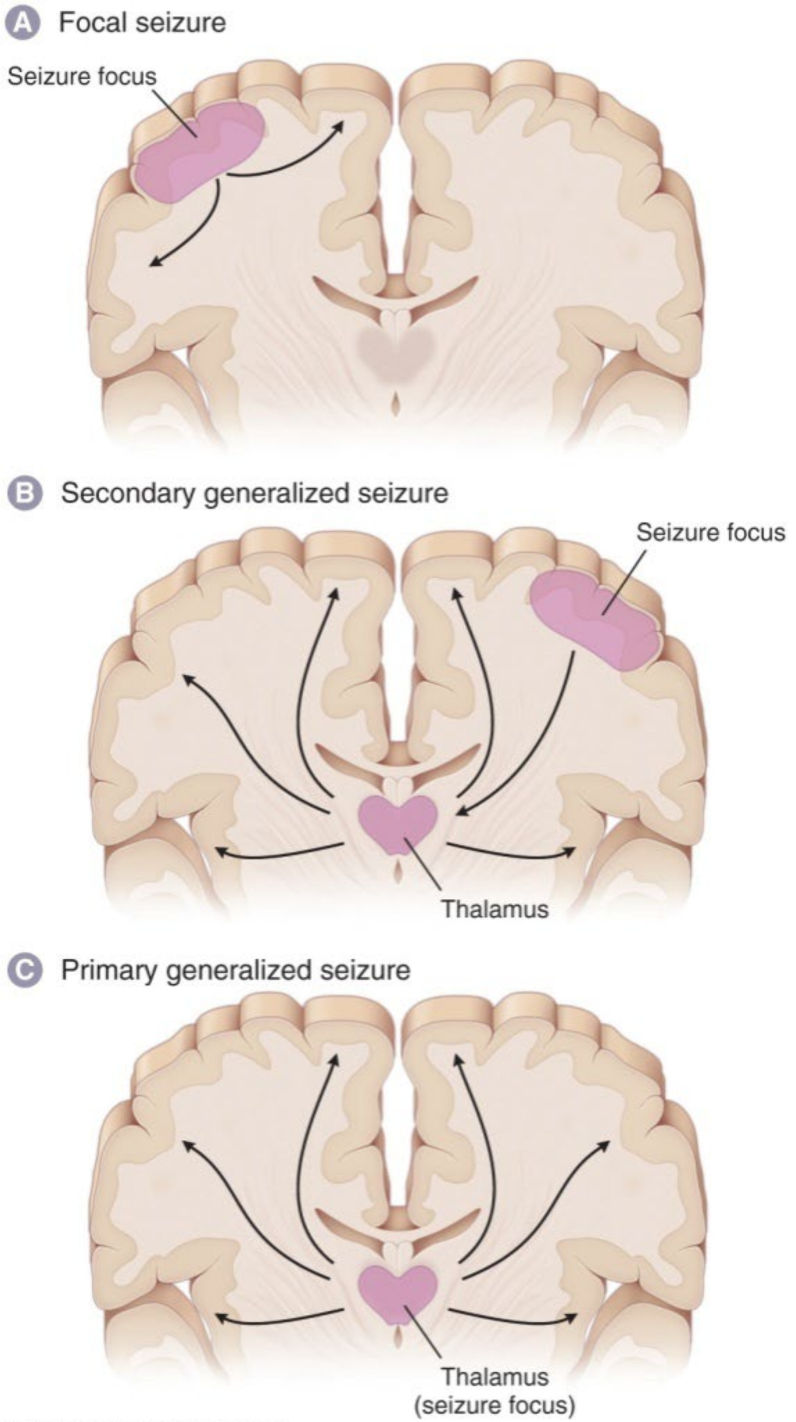
Objectives

- อธิบายกลไกการออกฤทธิ์และผลทางเภสัชวิทยาของ antiepileptics ได้
- เปรียบเทียบลักษณะทางเภสัชจลนศาสตร์ของ antiepileptics ที่มีความสำคัญทางคลินิกได้
- เปรียบเทียบความเสี่ยงของการเกิด type A ADRs ของ antiepileptics ที่สำคัญทางคลินิกได้
- ยกตัวอย่าง type B ADRs ของ antiepileptics ที่สำคัญได้
- เปรียบเทียบความเสี่ยงของการเกิดทารกวิรูปจากการใช้ antiepileptic ในระหว่างตั้งครรภ์ได้
- เปรียบเทียบความเสี่ยงของการเกิด drug interactions ของ antiepileptics ที่สำคัญทางคลินิกได้
- อธิบายผลทางเภสัชวิทยาจากคุณสมบัติการเป็น enzyme inducer ของ antiepileptics ได้

Epilepsy

- Seizure versus epilepsy
- Major types of epilepsy
  - Focal (partial) onset
    - Retained aware
    - Impaired awareness
  - Generalized onset
- Symptoms
  - Motor – tonic, clonic, atonic, myoclonic
  - Non-motor

Status epilepticus
Absence seizure – generalized non-motor
Dravet’s syndrome
Lennox–Gastaut syndrome



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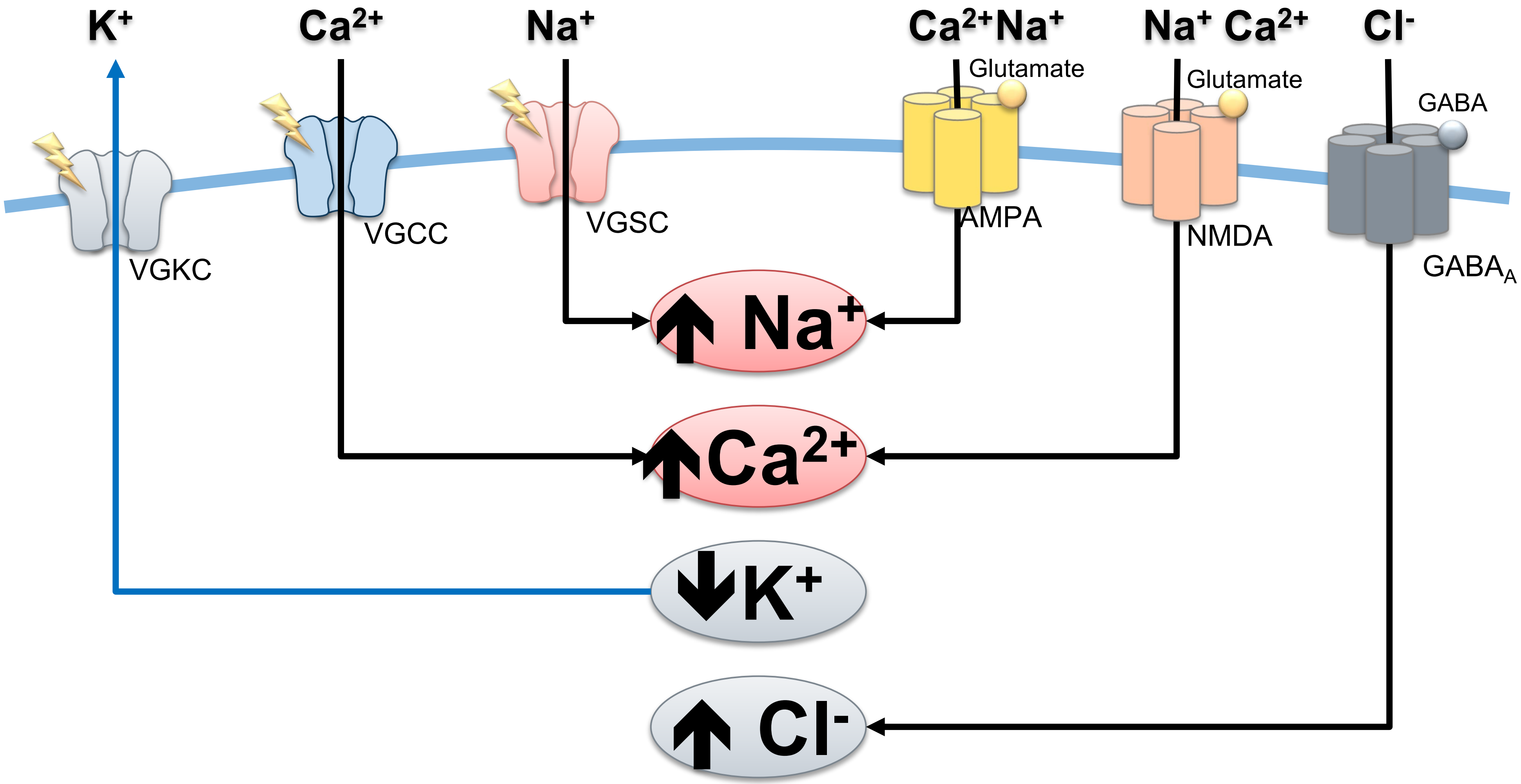
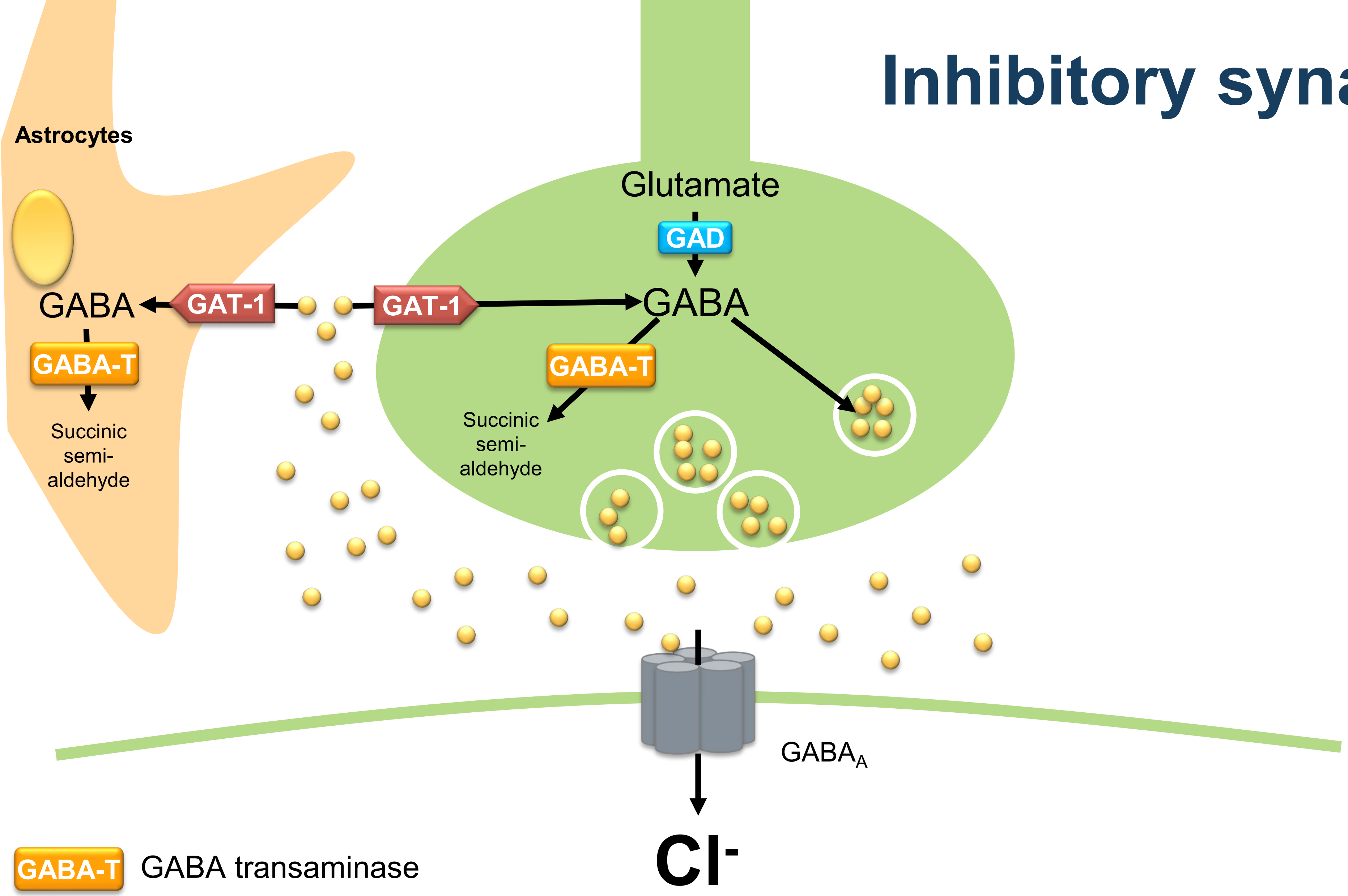
Controlling of neuronal excitability

Voltage-gated ion channels (sodium, calcium and potassium)

Ligand-gated ion channels (glutamate, GABA)



Inhibitory synapse

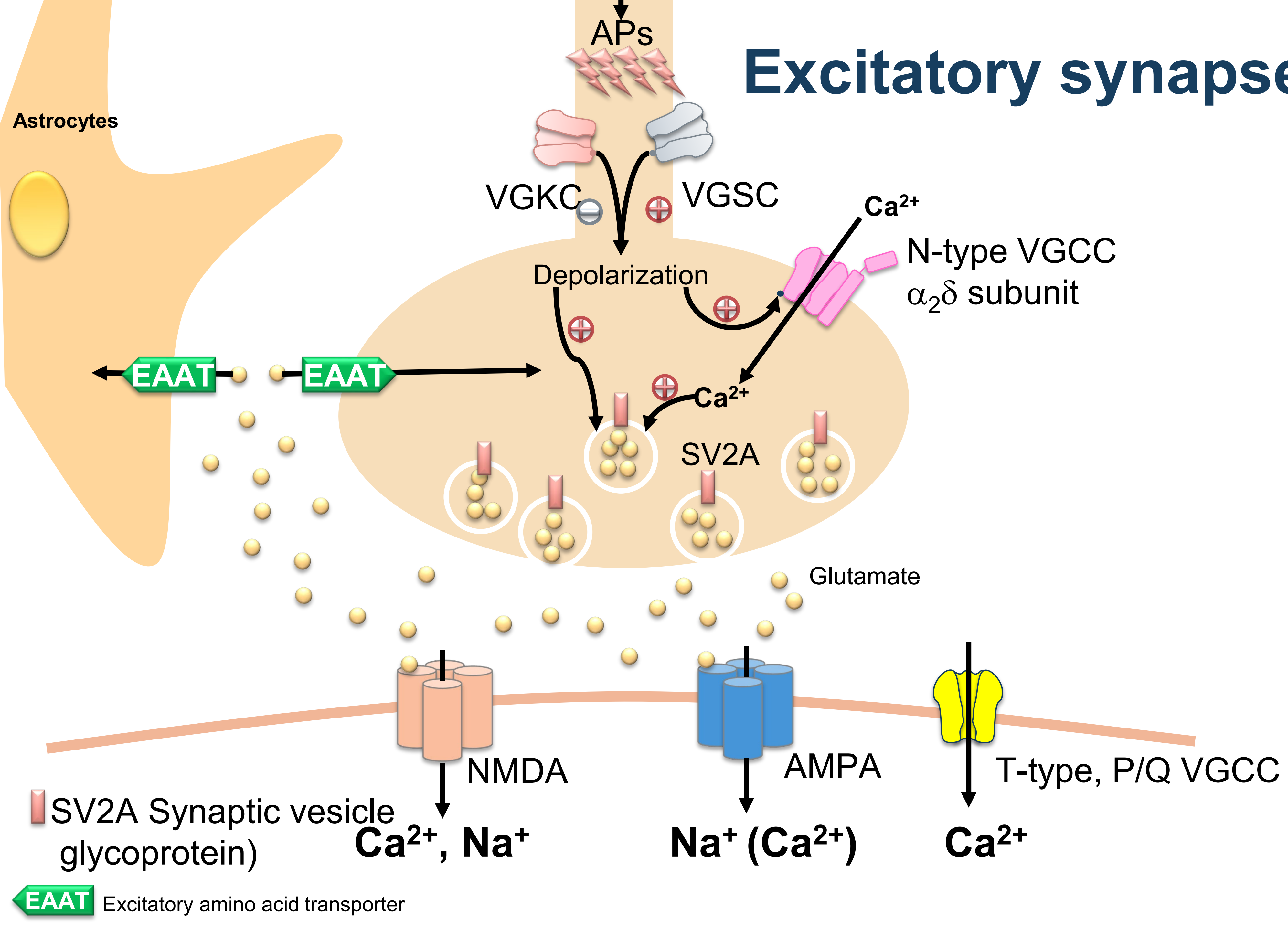


GABA-T GABA transaminase

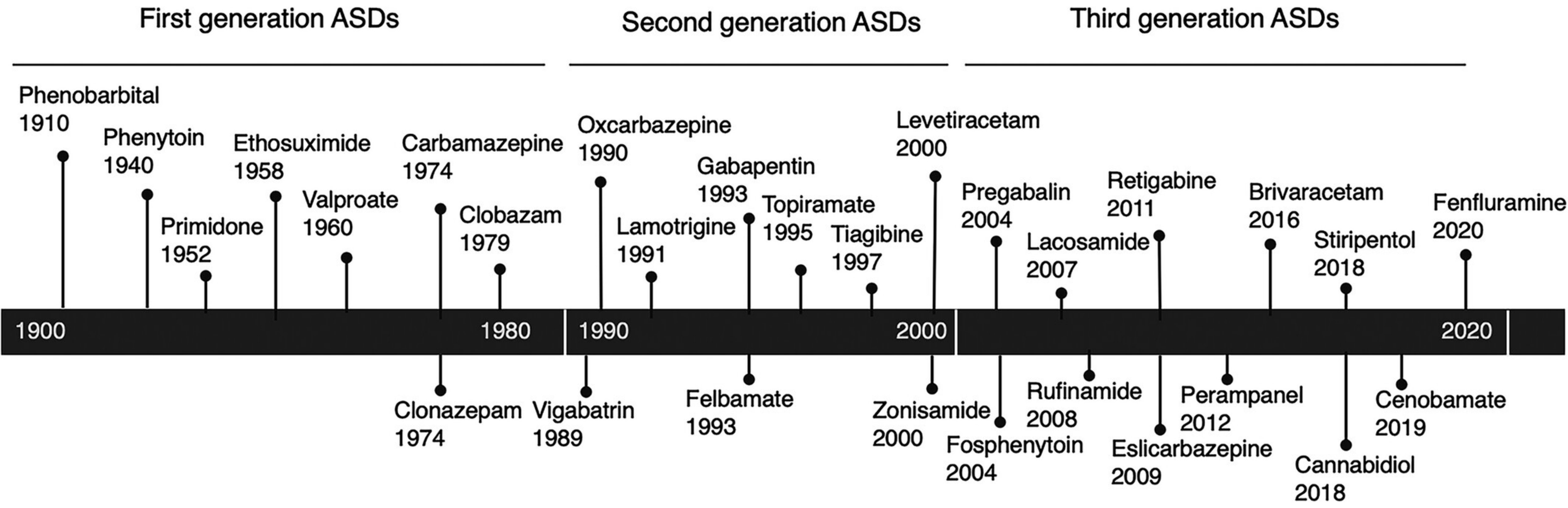
GAD Glutamic acid decarboxylase

GAT-1 GABA reuptake transporter -1

Excitatory synapse



Classification and year of introduction of antiseizure drugs (ASD)



Neuropsychopharmacol Rep. 2021 Sep;41(3):336-351.

Major AED classes based on main targets of action

- Sodium channel blockers
  - Phenytoin, fosphenytoin
  - Carbamazepine, oxcarbazepine, eslicarbazepine
  - Lamotrigine
  - Zonisamide, rufinamide, lacosamide
- Calcium channel blockers
  - N-type: Gabapentin, pregabalin, mirogabalin
  - T-type: Ethosuximide, valproate
- Synaptic vesicle glycoprotein 2A (SV2A) modulators
  - Levetiracetam
  - Brivaracetam
- AMPA blockers
  - Perampanel

- $GABA_A$ allosteric potentiating ligands (APLs)
  - Benzodiazepines, phenobarbital
- GABA turnover
  - GABA-T inhibitor: vigabatrin
  - GAT-1 inhibitor: tiagabine

- Broad spectrum (multimodal) AEDs
  - Sodium valproate (valproic acid)
  - Topiramate, felbamate, cenobamate
  - Zonisamide, rufinamide
  - Cannabidiol

AED	VGSC	VGCC HVA (N,P/Q, R, L)	VGCC LVA channels (T)	VGKC	SV2A	Glutamate receptors	$GABA_A$ receptor	GABA turnover	CAI
Benzodiazepines							+++		
Carbamazepine	+++								
Ethosuximide			+++						
Phenobarbital		+				+	+++		
Phenytoin	+++								
Valproic acid	++		++				++	++	
Felbamate	++	++				++ NMDA	++		
Gabapentin		++						+	
Lamotrigine	+++	++							
Levetiracetam		+			+++		+		
Oxcarbazepine	+++	++							
Pregabalin		++							
Tiagabine								+++	
Topiramate	++	++				++ AMPA	++		+
Vigabatrin							+	+++	
Zonisamide	+++		++						+
Eslicarbazepine	+++								
Lacosamide	+++ Slow								+
Perampanel						+++ $AMPA^X$			
Retigabine				+++ Activate			+		
Rufinamide	+++								

+ + + = principal target, + + = probable target, + = possible target https://epilepsysociety.org.uk/sites/default/files/2020-08/Chapter25Sills2015.pdf ^{X}non-competitive

Example of antiepileptic available in Thailand

Generic name	Trade name	Preparations
Carbamazepine	Tegretol®, Tegretol CR®	Tegretol tablet 200 mg, syr 100 mg/5 mL Tegretol CR tablet 200, 400 mg
Phenytoin	Dilantin®	Kapseal 100 mg, Infatab 50 mg, solution for injection 50 mg/mL
Fosphenytoin	Cereneu®	Solution for injection 75 mg/ ml (50 mg phenytoin equivalent)
Rufinamide	Inovelon®	Tablet 200 mg
Lacosamide	Vimpat®	Tablet 50, 100 mg, solution for infusion 10 mg/mL
Zonisamide	Zonigran®	Tablet 100 mg
Lamotrigine	Lamictal®	Tablet 25, 50, 100 mg
Oxcarbazepine	Trileptal®	Tablet 300, 600 mg
Phenobarbital	--	Tablet 32.5, 65 mg
Clobazam	Frisium®	Tablet 5 mg
Gabapentin	Neurontin®	Capsule 100, 300, 400 mg, tablet 600 mg
Pregabalin	Lyrica®	Capsule 25, 75, 150 mg
Perampanel	Fycompa®	Tablet 2, 4, 8 mg
Levetiracetam	Keppra®	Tablet 250, 500 mg, solution for injection 100 mg/mL, solution for injection 500 mg/5 mL
Topiramate	Topamax®	Tablet 25, 50, 100 mg
Sodium valproate	Depakine, Depakine Chrono®	Depakine Chrono CR tablet 500 mg Depakine enteric-coated tablet 200 mg, oral solution 200 mg/mL, powder for injection 400 mg/4 mL

$GABA_A$ allosteric potentiating ligands (APLs)

Benzodiazepines, clonazepam, clobazam, diazepam, midazolam
Phenobarbital

Phenobarbital

- PK
  - Metabolized by CYP2C9/2C19
  - 25% excreted unchanged in urine
- Strong enzyme inducer
- ADR
  - Sedation, respiratory depression
  - Rash and other idiosyncratic reactions
  - Reduced IQ in long term use, may induce hyperactive child

Skin reactions induced by AEDs

Maculopapular rash

Toxic epidermal necrolysis (TEN)

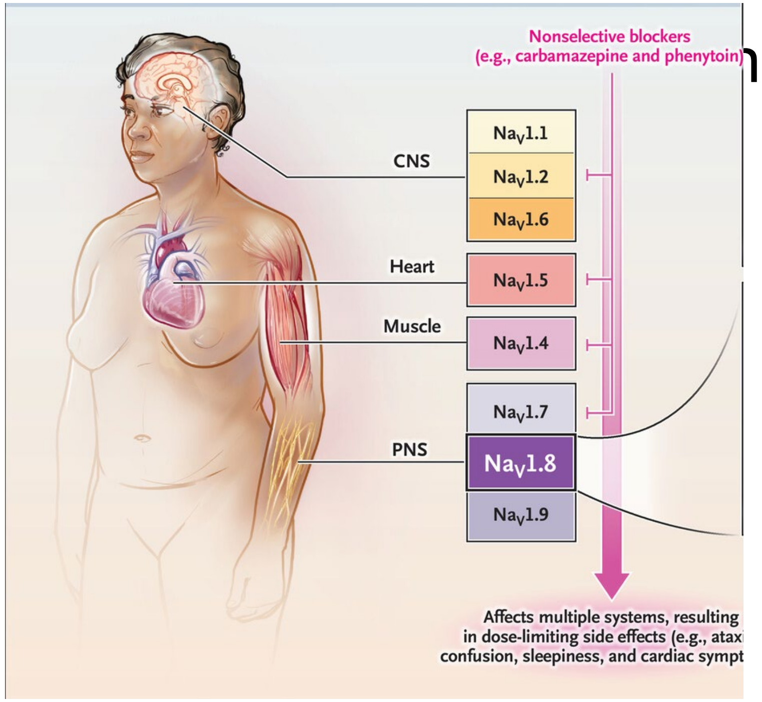
Stevens-Johnson Syndrome (SJS)

Sodium Channel Blockers

Generic name	Trade name	Preparations
Carbamazepine	Tegretol®, Tegretol CR®	Tegretol tablet 200 mg, syr 100 mg/5 mL Tegretol CR tablet 200, 400 mg
Oxcarbazepine	Trileptal®	Tablet 300, 600 mg
Phenytoin	Dilantin®	Kapseal 100 mg, Infatab 50 mg, solution for injection 50 mg/mL
Fosphenytoin	Cereneu®	Solution for injection 75 mg/ ml (50 mg phenytoin equivalent)
Rufinamide	Inovelon®	Tablet 200 mg
Lacosamide	Vimpat®	Tablet 50, 100 mg, solution for infusion 10 mg/mL
Zonisamide	Zonigran®	Tablet 100 mg
Lamotrigine	Lamictal®	Tablet 25, 50, 100 mg

Classification of Na_V

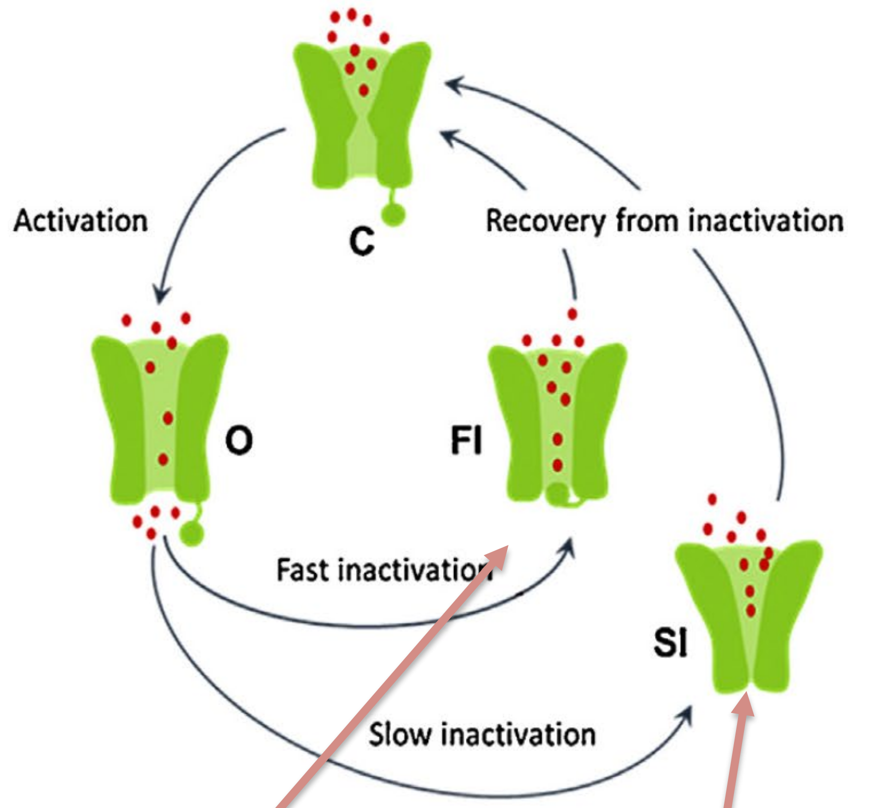
- By amino acid homology
  - Na_V1.1 and 1.2 - brain
  - Na_V1.3, 1.7, 1.8, 1.9 peripheral nerve
  - Na_V1.4 skeletal muscle
  - Na_V1.5 cardiac tissue
  - Na_V1.6 brain, cerebellum
- By inactivation states
  - Fast inactivation state
  - n state



Mechanisms

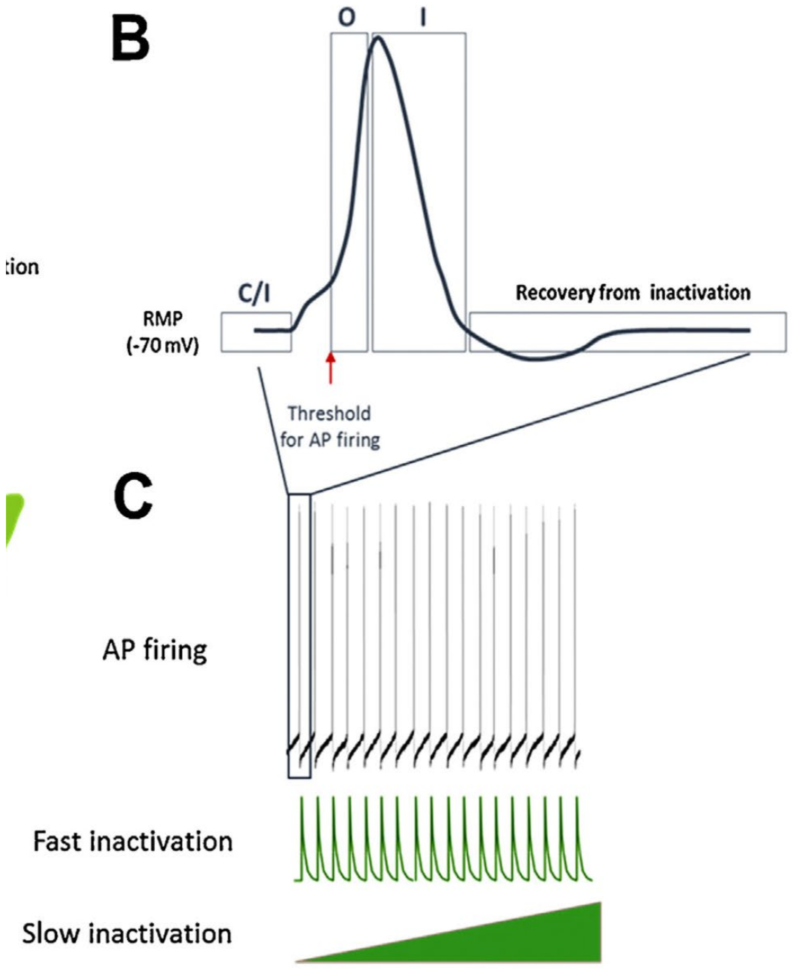
- Reversible blockers of Na_V channels
- Most are non-selective to Na_V subtypes
- Most VGSC blockers AEDs- high affinity to fast inactivation state
- Lacosamide - high affinity to slow inactivation state

VGSC working cycles

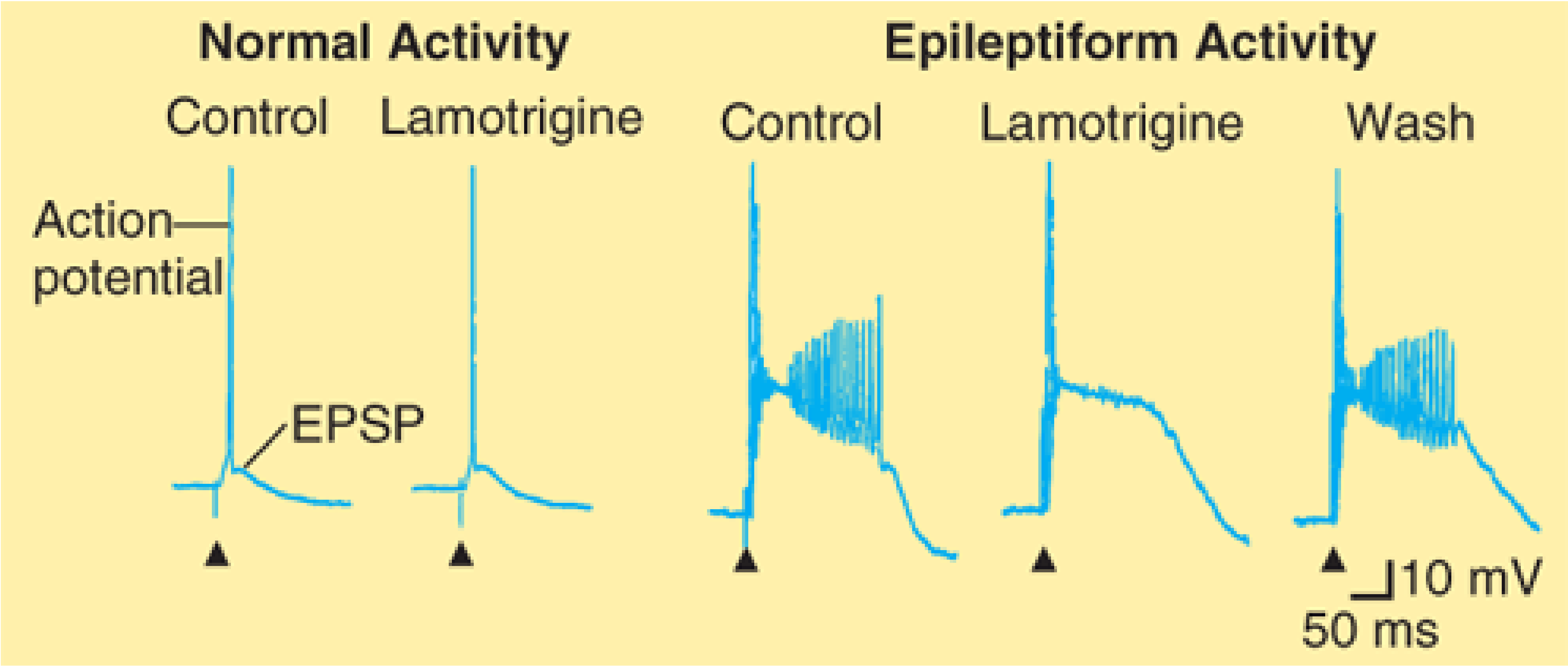


Phenytoin
Carbamazepine
Oxcarbazepine
Eslicarbazepine acetate
Lamotrigine
Rufinamide
Topiramate
Valproate
Zonisamide

Lacosamide



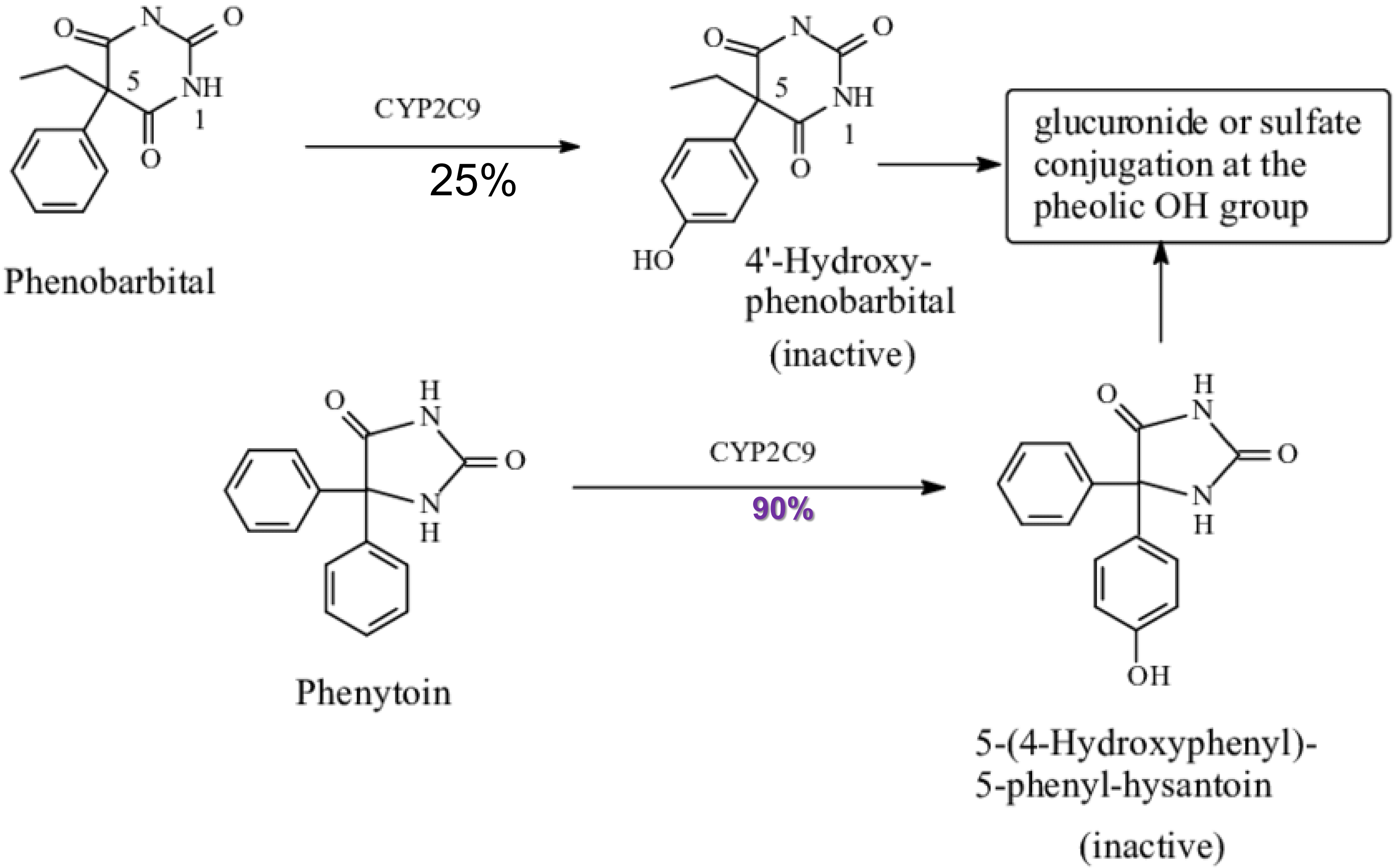
Use-dependent blockade by lamotrigine



Phenytoin

- Absorption is variable and incomplete
- High protein bound 90%
- Metabolized 95% in liver by CYP2C9 and CYP2C19
  - Saturable
  - Nonlinear pharmacokinetics
  - Variability
    - CYP2C9 polymorphism
    - Rate increased in younger children, pregnancy, during mens, acute trauma
    - Rate decreased in advanced age
- Strong enzyme inducer

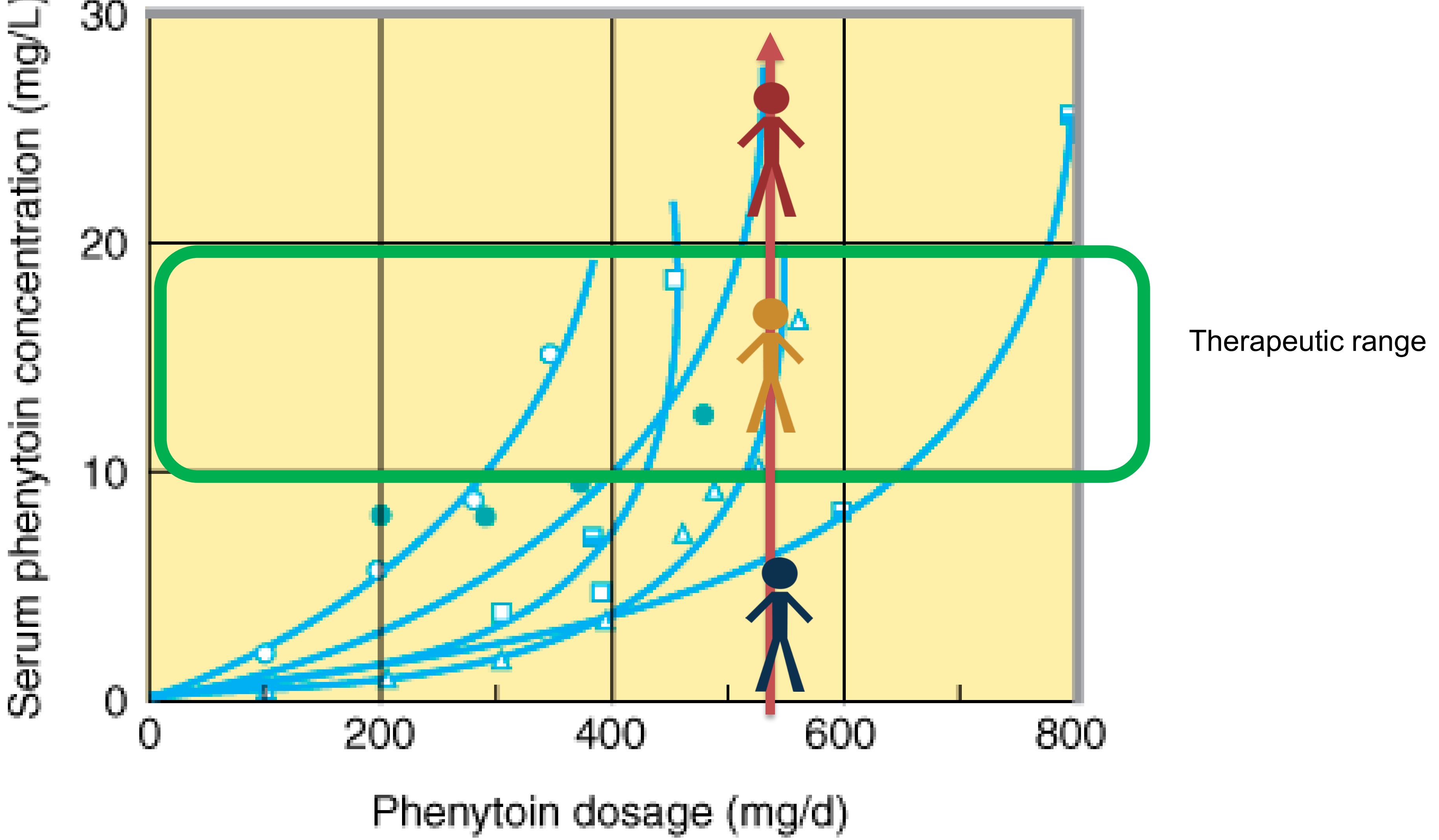
Metabolism pathway of phenobarbital and phenytoin (by CYP2C9) – saturable metabolism



Individual variability in phenytoin plasma level

- Dose – saturation of metabolism
- CYP2C9 polymorphism

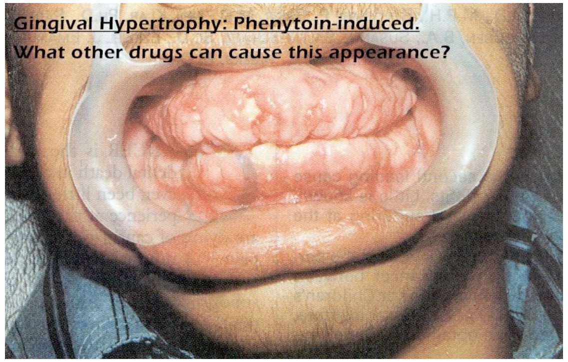
Pharmacokinetics of phenytoin is highly variable and non-linear



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

ADRs of phenytoin

- Dose-related CNS side effects
  - Dizziness, sedation, ataxia, nystagmus, diplopia
- Arrhythmia if iv infusion rate too high
- Gingival hyperplasia, hirsutism, oily skin, acne
- Rash and idiosyncratic reactions
- Phenytoin-induced purple glove syndrome (PGS)
  - Peripheral IV administration of phenytoin



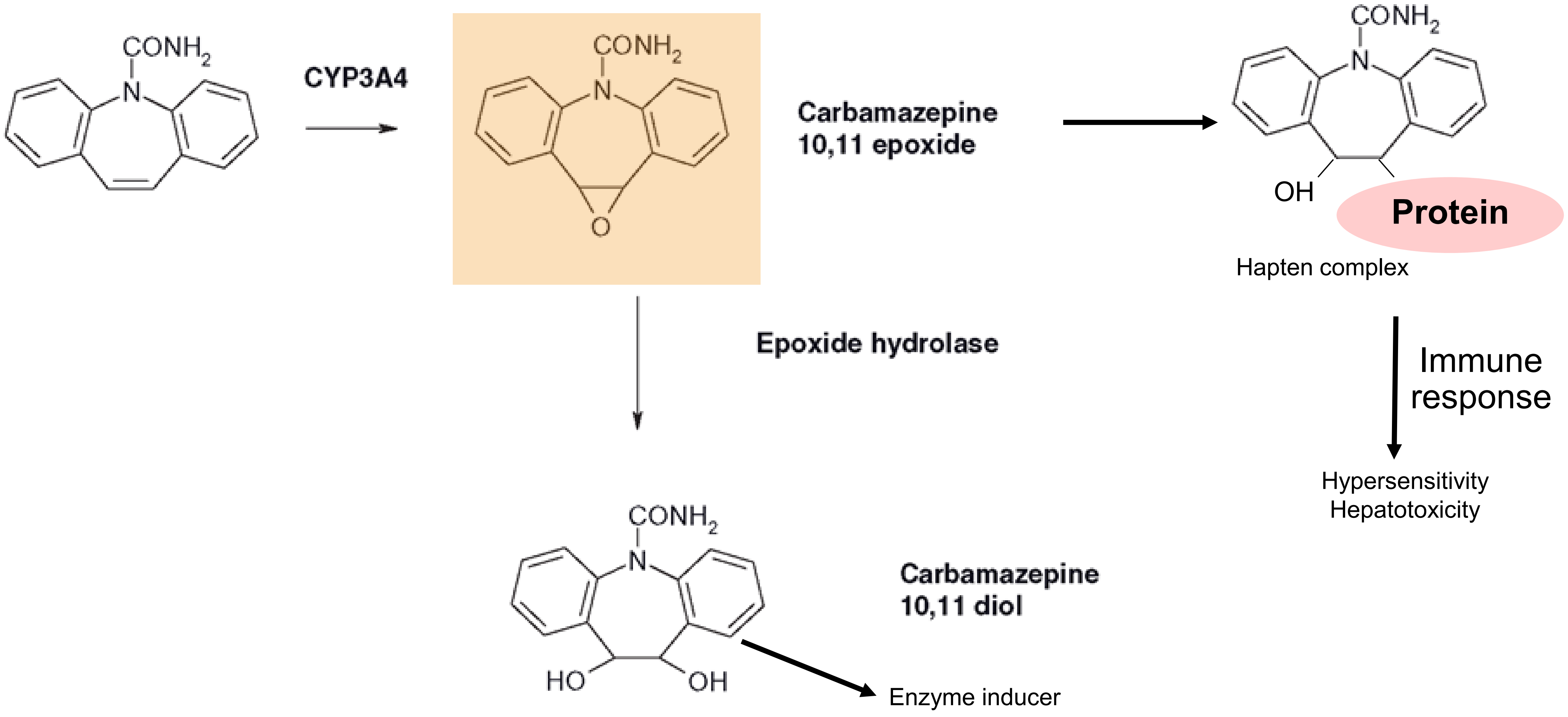
Gingival hyperplasia (Lamey, 1988)

FIG 2. Left (a) and right (b) hands of an 81-year-old woman after 15 days of intravenous phenytoin administration (100 mg every 8 hours)

Carbamazepine

- Metabolism 98% in liver by CYP3A4
- Potent CYP3A4 enzyme inducer and also induce other enzymes
- Autoinduction
- Time-dependent PK
  - First month half-life is 30 hr and then shortened to 12-24 hr

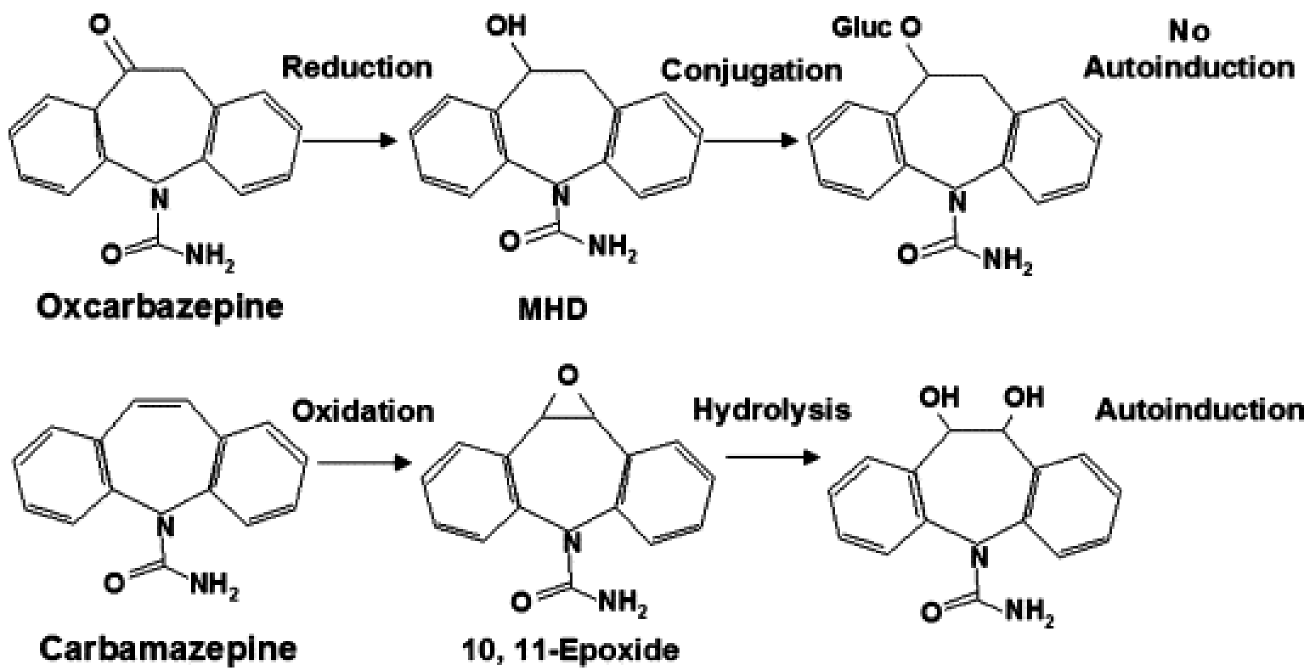
Metabolism of carbamazepine



ADR of carbamazepine

- Dose-related CNS side effects similar to phenytoin but more sedated
- Abnormal liver enzyme is common (usually transient) but may develop fatal hepatitis
- *HLAB*1502* positive greatly increases the risk of developing serious cutaneous skin reactions e.g. Stevens-Johnson syndrome and toxic epidermal necrolysis
- Hyponatremia – augmentation of ADH

Oxcarbazepine metabolism



- Less induction of CYP enzyme
- Less risk of rash than CBZ and milder in severity
  - CBZ 7%
  - OXC 4-5%
- Cross-reactivity when switching CBZ to OXC 25-30%

Lamotrigine

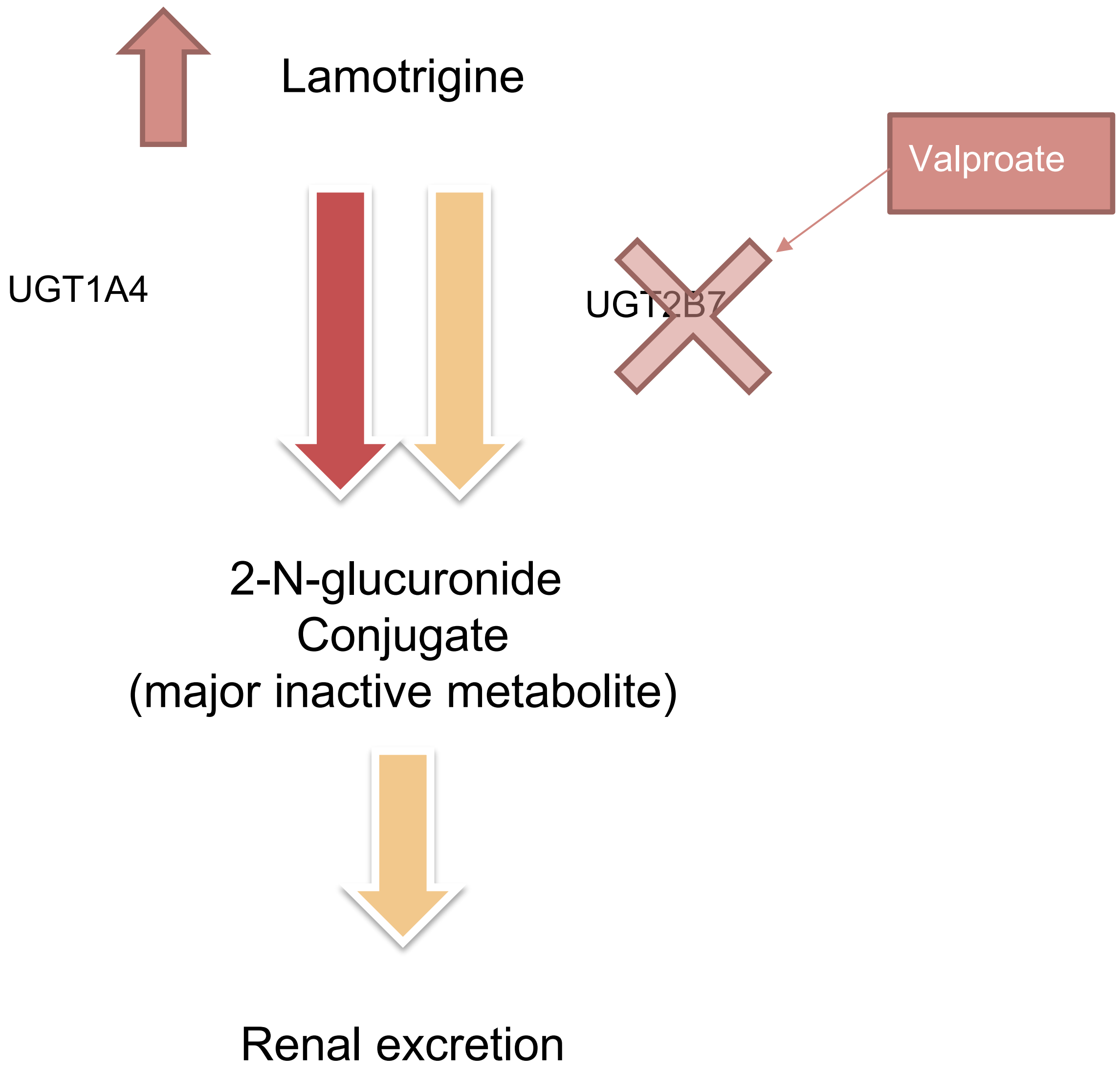
- Multiple mechanisms of action beneficial in mood disorders
  - Block VGSC, VGCC - antiepileptic and anti-manic effects
  - 5-HT_{1A} agonist –antidepressant effects
  - Block monoamine reuptake (high dose) –antidepressant effects

- PK
  - Metabolism by UDP-glucuronosyl transferase

ADRs of lamotrigine

- Less sedated
- Serious rash
  - Dose dependent
  - Risk of rash is higher in
    - Pediatric patients
    - Coadministration of valproate
  - Exceeding the recommended dose

Lamotrigine needs to be initiated with low dose and titrated slowly to prevent rash.


Lamotrigine
Valproate
UGT1A4
UGT2B7
2-N-glucuronide
Conjugate
(major inactive metabolite)
Renal excretion

Calcium channel blockers

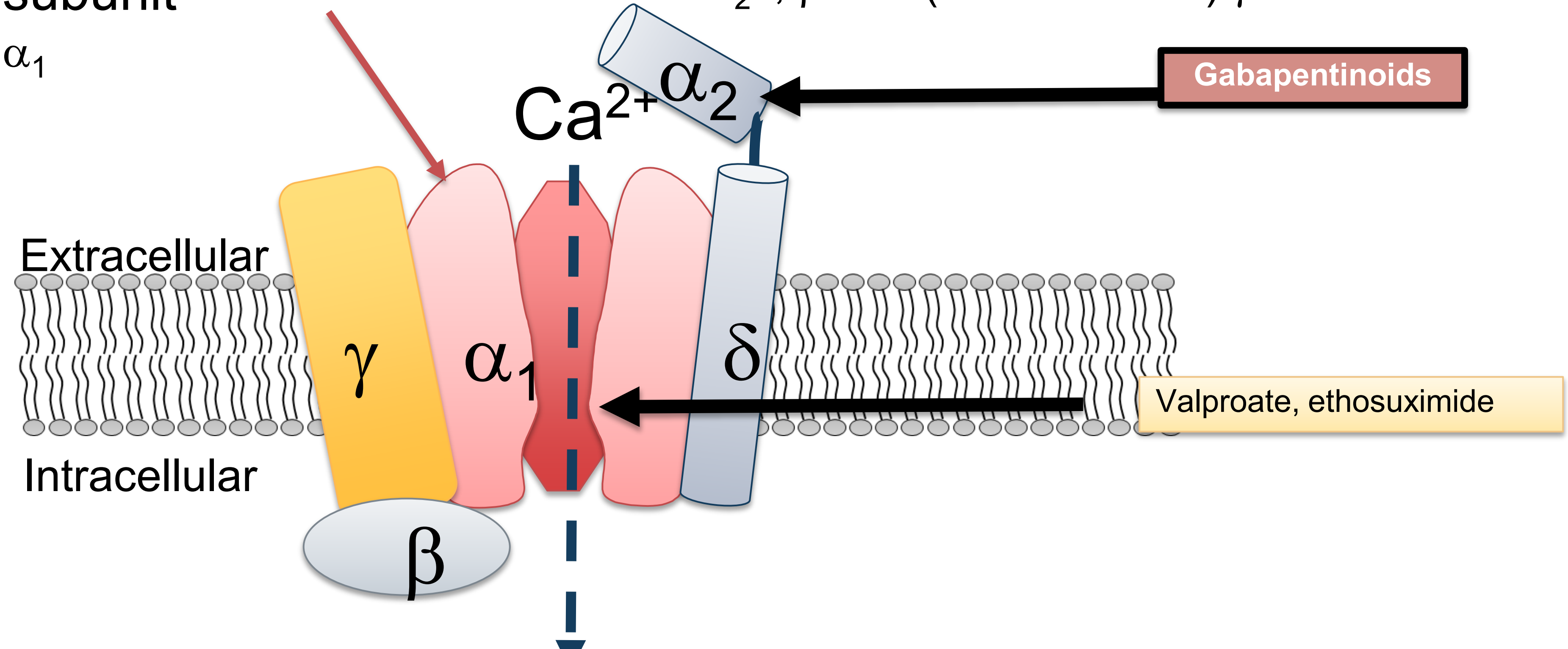
Generic name	Trade name	Preparations
Gabapentin	Neurontin®	Capsule 100, 300, 400 mg, tablet 600 mg
Pregabalin	Lyrica®	Capsule 25, 75, 150 mg
Mirogabalin (not for epilepsy)	Tarlige®	Tablet 2.5, 5, 10, 15 mg

Molecular structure of VGCC

L-type
P/Q-type
N-type
R-type
T-type

Pore forming subunit
α_1

Auxilary subunits
$\alpha_2\delta$, β and (in some case) γ



- Neuronal excitability
- Activation of Ca^{2+}-dependent enzymes e.g. protein kinase, MAPK, phospholipases
- Neurite growth
- Neurotransmitter release
- Central sensitization

Calcium channel subtypes and localization

Subtypes			Localization
Alias	Channel	Current	
α_{1S}	$Ca_V1.1$	L-type	Skeletal muscle
α_{1C}	$Ca_V1.2$		Heart, brain, smooth muscle, adrenal, kidney
α_{1D}	$Ca_V1.3$		Brain, kidney, pancreas
α_{1F}	$Ca_V1.4$		Retina
α_{1A}	$Ca_V2.1$	P/Q-type	Brain, pituitary, kidney
α_{1B}	$Ca_V2.2$	N-type	Brain, nervous system
α_{1E}	$Ca_V2.3$	R-type	Brain, heart, pituitary
α_{1H}	$Ca_V3.1$	T-type	Brain, kidney, nervous system
α_{1H}	$Ca_V3.2$		Brain, heart, kidney, liver
α_{1I}	$Ca_V3.3$		Brain

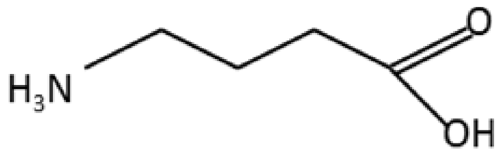
Gabapentinoids

Valproate, ethosuximide

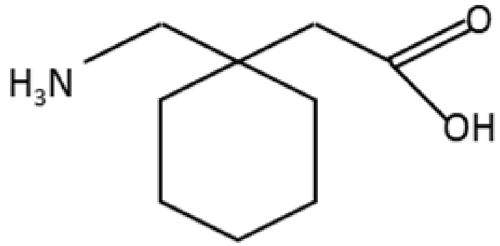
Mol Interv 2004;4:116-18.

Chemical structures of GABA and gabapentinoids

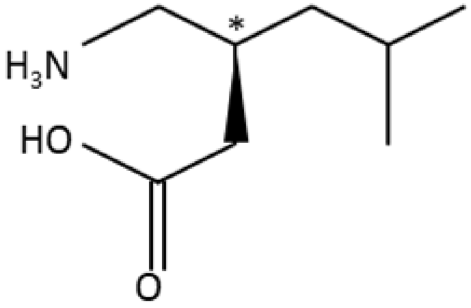
GABA



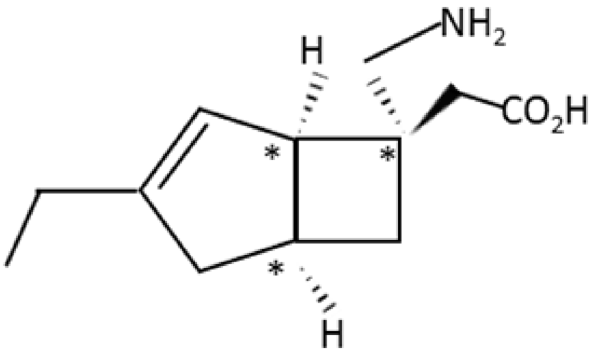
Gabapentin



Pregabalin



Mirogabalin

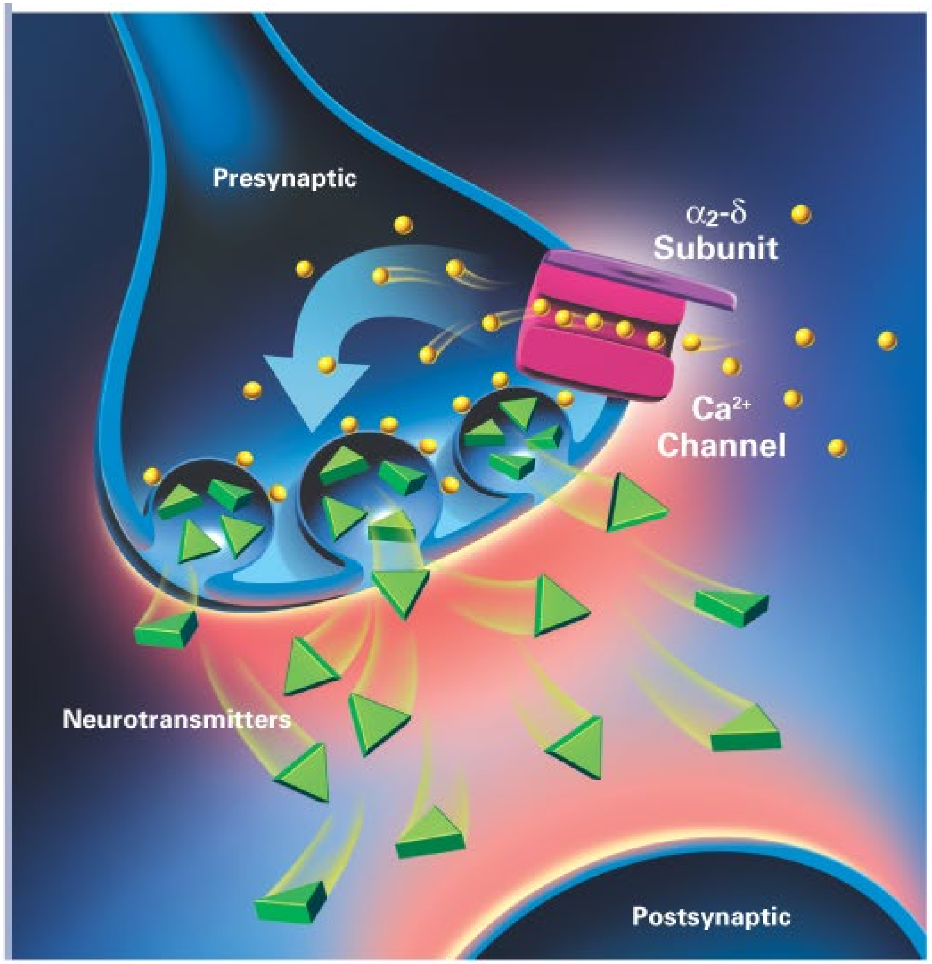


Gabapentinoids

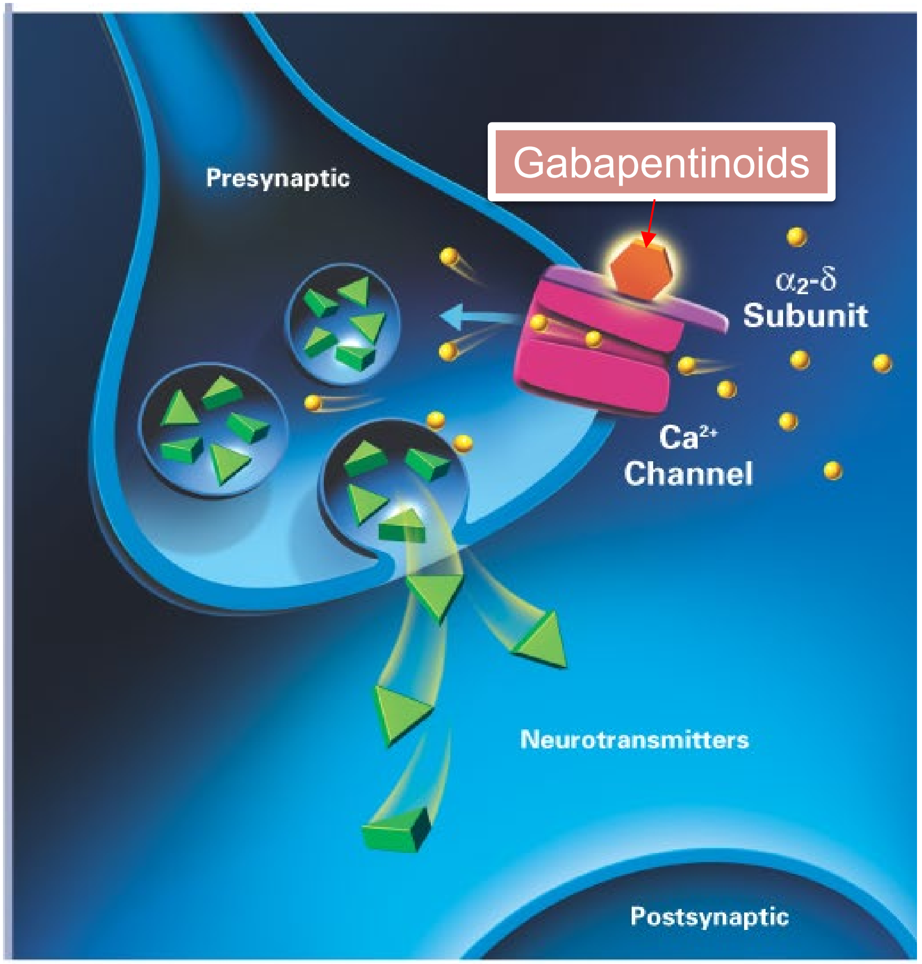
- Mechanisms
  - Bind to $\alpha_2\delta$ subunit and then block the N-type calcium channel, so call $\alpha_2\delta$ ligands
- Some other antiepileptics block at the ion pore of the channel
  - Valproate and ethosuximide -> T-type calcium channel
  - Topiramate -> N-type calcium channel

Gabapentinoids modulate hyperexcited neurons

Hyperexcited neuron



Hyperexcited neurons modulation by gabapentinoids

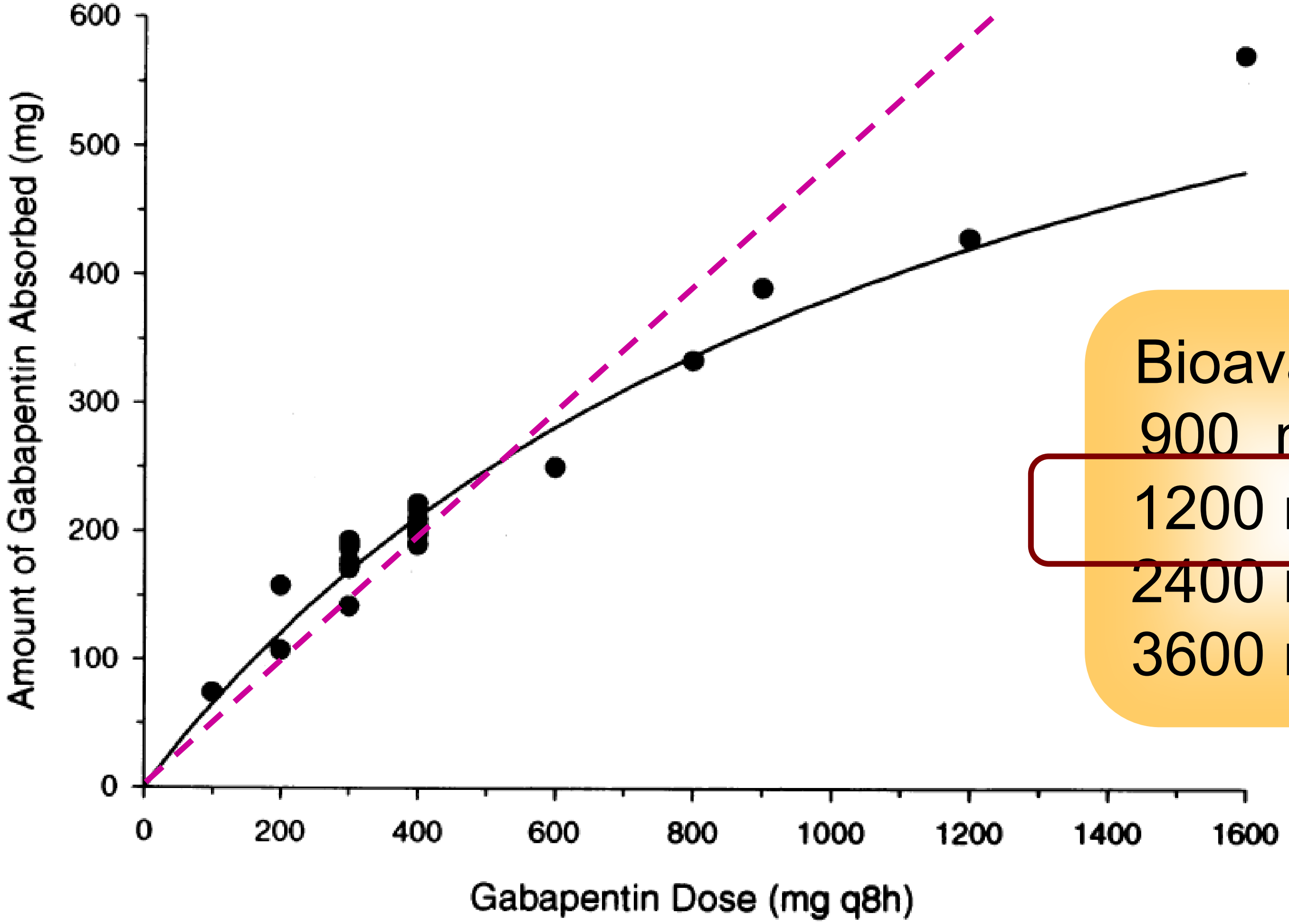


Modified from Eur Neuropsychopharmacol 2006;16 suppl 2:128-33.

Differences between gabapentinoids

	Gabapentin	Pregabalin	Mirogabalin
Absorption	Saturable	Non-saturable across dose range	Non-saturable across dose range
Tmax	2-3 hr	0.8-1.4 hr	0.5-2hr
Elimination T1/2	5-7 hr	5-6.5 hr	2-3 hr
Protein binding	3%	None	25%
Metabolism	Not metabolized	0.9% to N-methylated derivative	Negligible
Renal excretion	76% to 81%	90% to 99%	96% (76.4% unchanged)
Starting dose	100-900 mg/day Dose reduction required with renal insufficiency	75-150 mg/day Dose reduction required with renal insufficiency	5 mg Dose reduction required with renal insufficiency
Dosing frequency	Every 8 hr	Every 8-12 hr	Every 12-24 hr
Usual effective dose	1200-2400 mg	150-600 mg	?
Max. dose	3600 mg (≤1200 mg/dose)	600 mg	30 mg

Observed (solid circles) and predicted (solid line) amount of gabapentin absorbed following a single oral dose of gabapentin to healthy volunteers



Bioavailability
900 mg 60%
1200 mg 47%
2400 mg 34%
3600 mg 33%

Epilepsy Research 1998; 31:91–99.

Other indications of gabapentinoids

- All are indicated for neuropathic pain
- Pregabalin is also indicated for generalized anxiety disorders and fibromyalgia
- Mirogabalin is not licensed as an antiepileptic

Summary of the mechanisms of drugs blocking voltage-gated ion channels

- Reversible blockers voltage-gated sodium channels
  - High affinity to fast inactivation state- other most VGSC blockers AEDs
  - High affinity to slow inactivation state- lacosamide

- Reversible blockers voltage-gated calcium channels
  - Bind to the auxillary subunit - gabapentinoids
  - Bind to the ion pore – valproate and ethosuximide

Broad Spectrum Antiepileptics

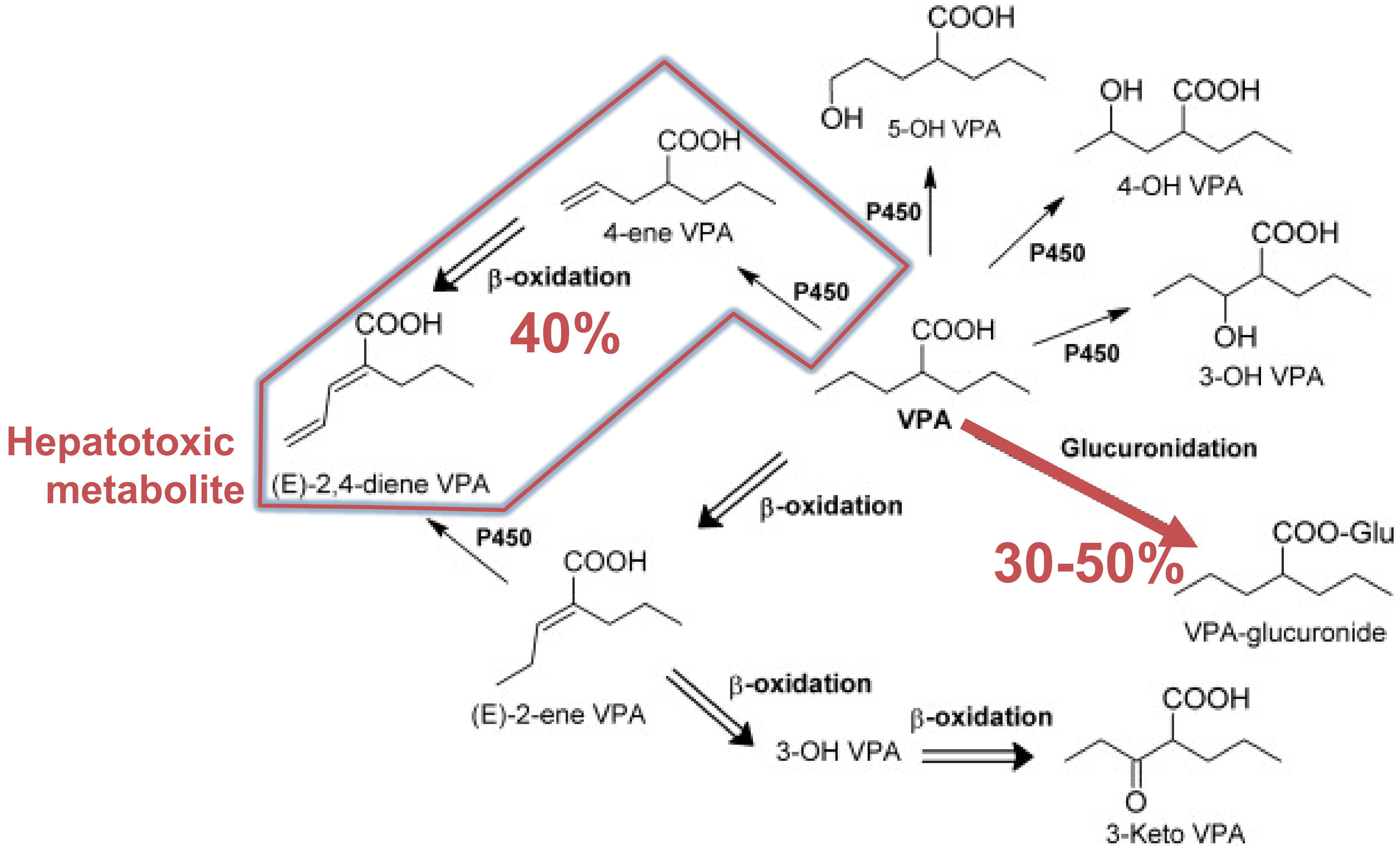
Generic name	Trade name	Preparations
Sodium valproate	Depakine, Depakine Chrono®	Depakine Chrono CR tablet 500 mg Depakine enteric-coated tablet 200 mg, oral solution 200 mg/mL, powder for injection 400 mg/4 mL
Topiramate	Topamax®	Tablet 25, 50, 100 mg

Valproate (Depakine®)

- Broad spectrum antiepileptic
  - Block VGSC
  - Block VGCC (T-type)
  - GABA APL
  - Inhibit GABA-T
- Effective in absence seizure
- Can be used in bipolar disorder, migraine prevention

- PK
  - High protein bound 90-95%
  - Metabolised in liver 95% by CYP450 and β-oxidation

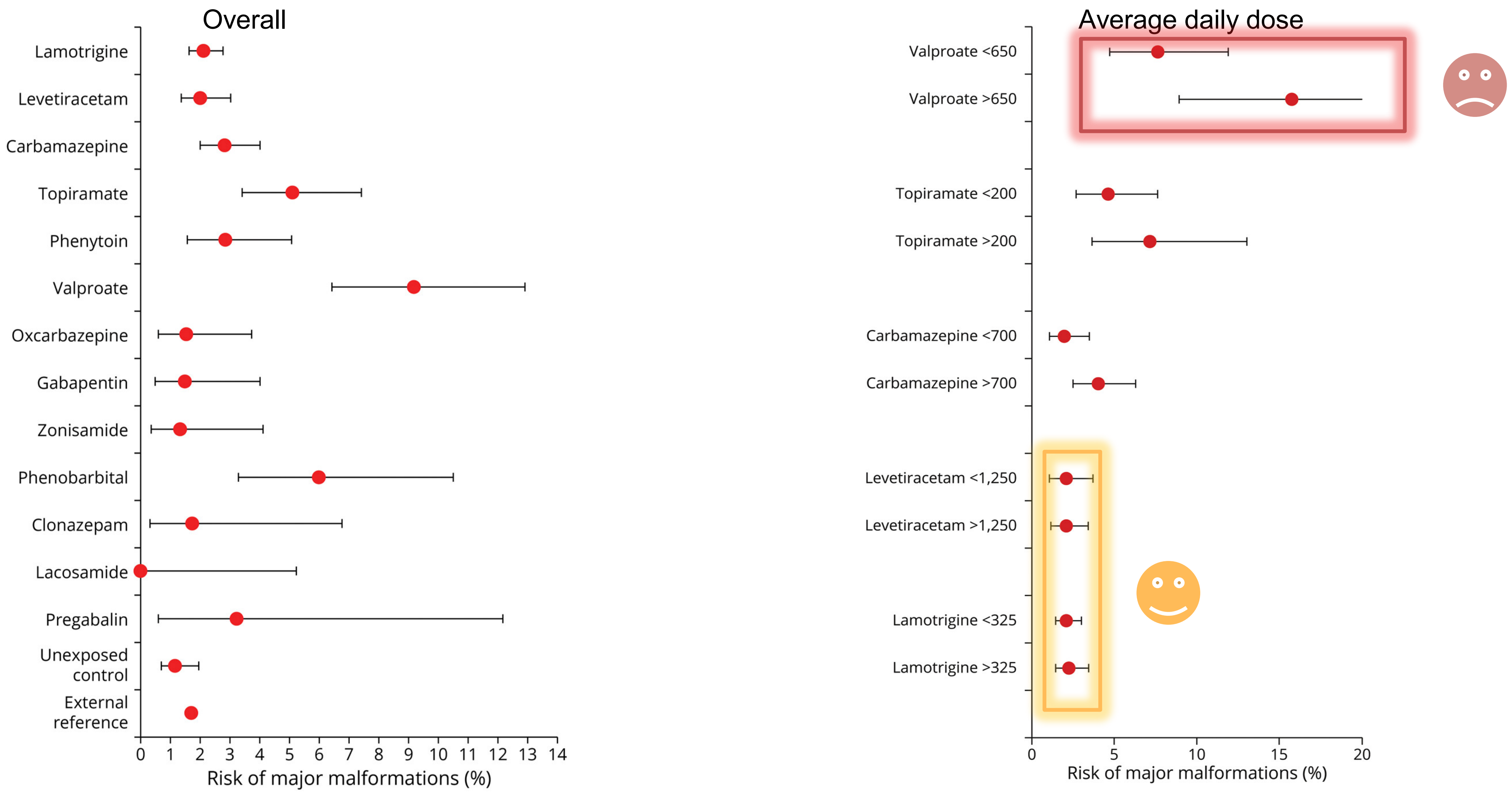
Valproate metabolism



ADRs of valproate

- Dose-related CNS side effects like other AEDs
- Tremor
- GI side effects in about 16%
  - Anorexia; nausea; vomiting, severe pancreatitis
- Weight gain 8-18%
- Alopecia 2-12%
- Thrombocytopenia, bleeding
- Hepatotoxicity, especially if <2 years of age
- Valproate-induced encephalopathy with hyperammonemia
- Teratogenicity - neural tube defects, major malformations $\approx$10%

Risk of major malformations in newborn from first trimester exposure of an ASM



Neurology. 2025 Aug 12;105(3):e213786. doi: 10.1212/WNL.0000000000213786. Epub 2025 Jul 16. PMID: 40669027.

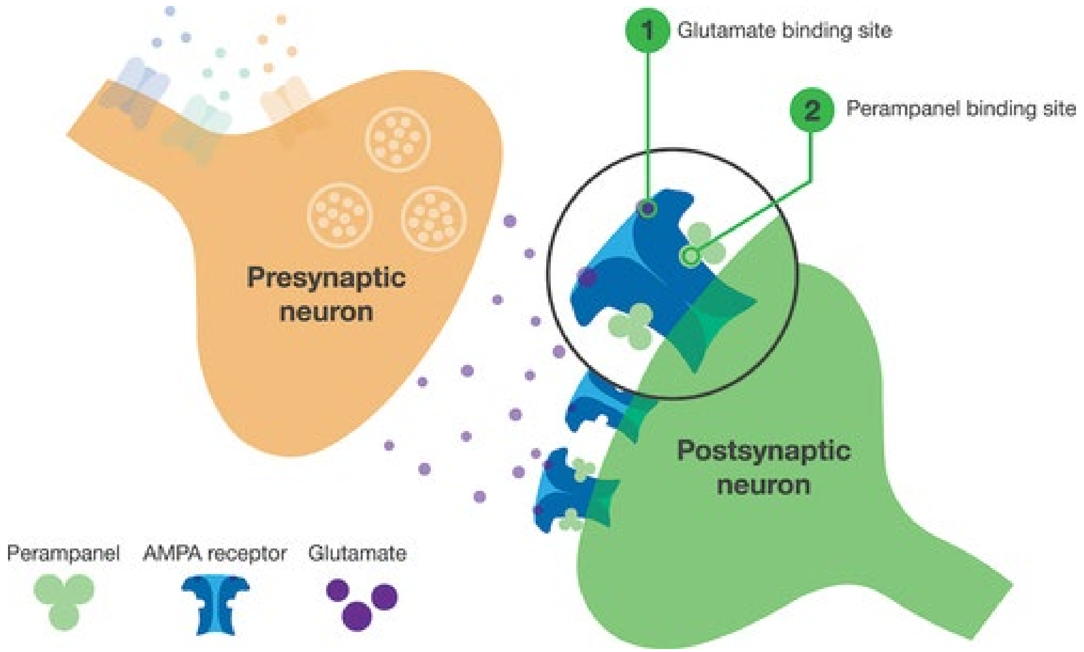
Topiramate (Topamax®)

- Broad spectrum antiepileptic
  - Block VGSC
  - Block VGCC
  - Block AMPA/kainate
  - $GABA_A$ APL
- Effective for migraine prevention

- PK
  - Low protein bound 15-41%
  - Primarily (70%) eliminated unchanged in the urine
- ADRs
  - Cognitive impairment, slow thinking
  - Paresthesia
  - Weight loss
  - Hypohydrosis
  - Kidney stones

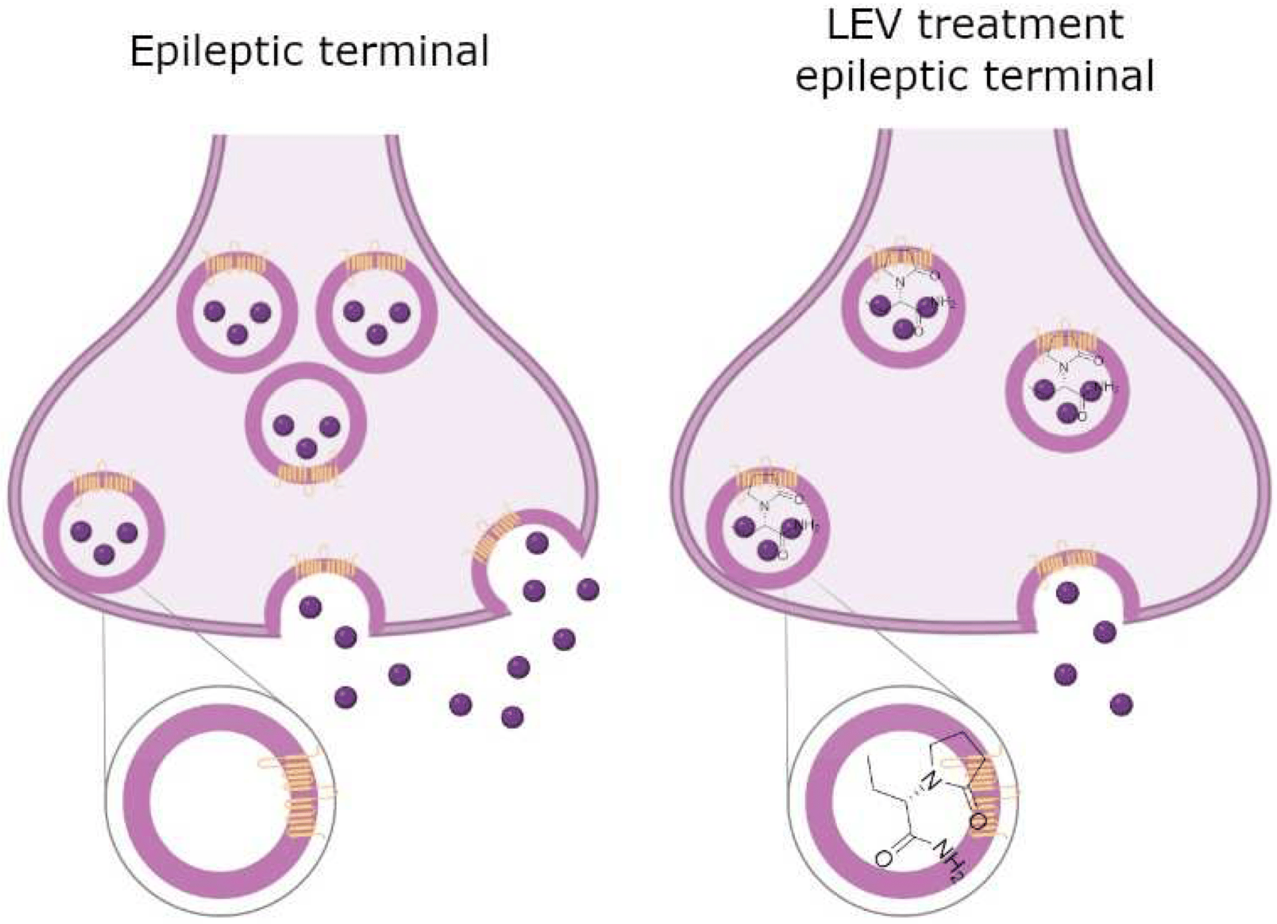
Perampanel (Fycompa®)

- Mechanism
  - Non-competitive AMPA receptor blocker
- PK
  - Mainly metabolized by CYP3A4
- ADRs
  - Aggression, irritability, psychosis, suicidal ideation

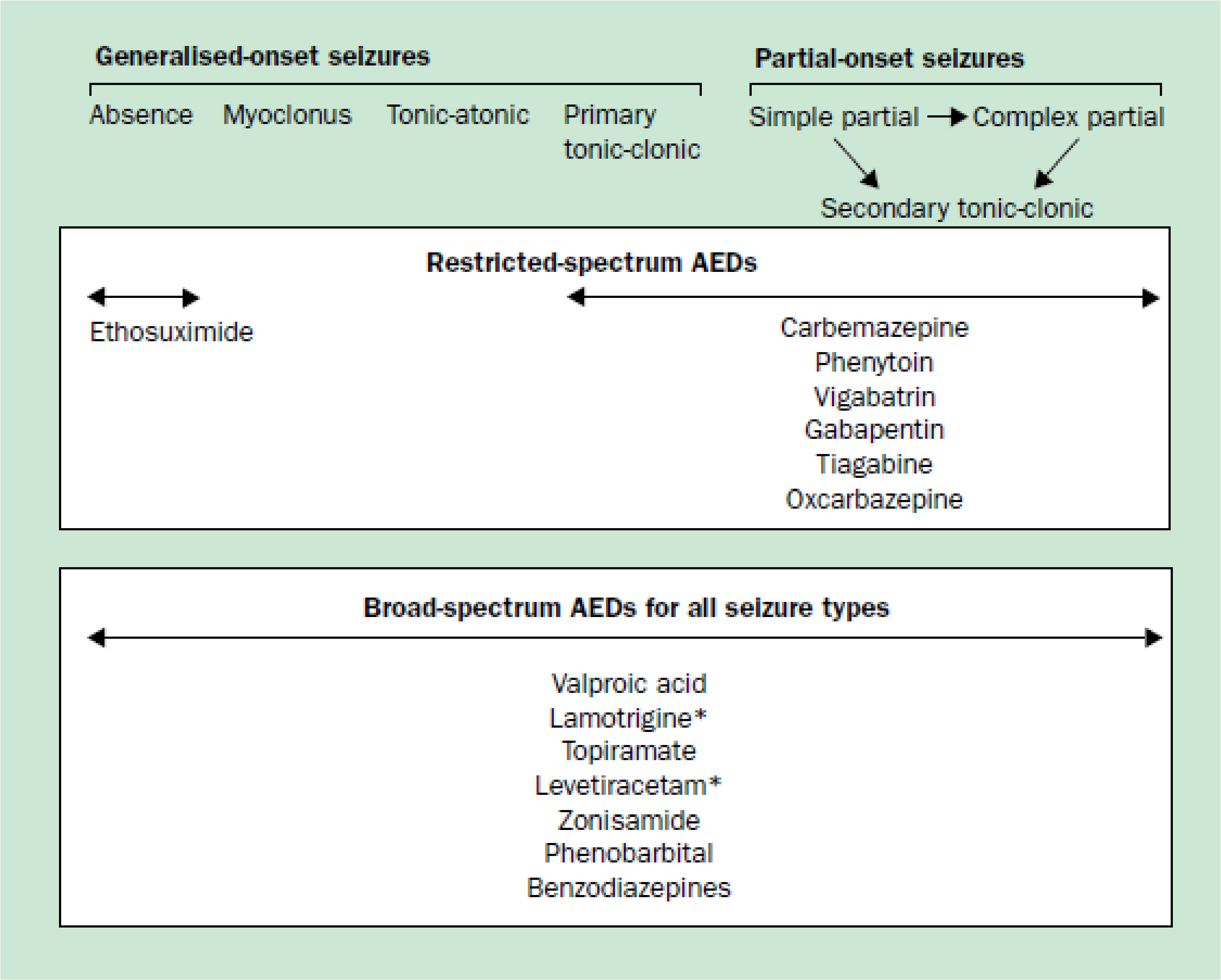


Levetiracetam (Keppra®)

- Mechanism
  - Binding to SV2A and inhibit glutamate release
- Preferentially act at excitatory synapse
- PK
  - Mainly excreted unchanged by kidney
- ADR
  - Aggression, irritability, psychosis



Spectrum of antiepileptics



Lancet Neurol 2004;3:729-35.

ชนิดของการชัก	บัญชียา ก	บัญชียา ข	บัญชียา ค	ไม่อยู่ในบัญชียาหลักแห่งชาติ
Adults with partial onset seizure	carbamazepine phenytoin sodium valproate phenobarbital	clonazepam	lamotrigine (elderly) topiramate levetiracetam gabapentin (elderly)	oxcarbazepine zonisamide clobazam pregabalin
Children with partial onset seizure	carbamazepine phenytoin phenobarbital sodium valproate	clonazepam	topiramate lamotrigine	oxcarbazepine zonisamide clobazam
Generalized tonic clonic seizure	phenobarbital sodium valproate phenytoin carbamazepine	clonazepam	lamotrigine topiramate levetiracetam gabapentin	oxcarbazepine clobazam
Absence epilepsy	sodium valproate	clonazepam	lamotrigine	
Juvenile myoclonic epilepsy	sodium valproate		topiramate	
Atonic/tonic seizure	sodium valproate	clonazepam	topiramate lamotrigine nitrazepam levetiracetam	

Thai CPG of Epilepsy 2559.

Adverse Drug Reactions of Antiepileptics

ADRs of commonly used antiepileptic drugs

- General CNS side effects
  - Dizziness, fatigue, sedation
  - Diplopia, blurred vision, nystagmus, ataxia
  - Cognitive disturbance
    - Memory problem, slow thinking
    - Lower IQ in long-term use
  - Weight change
- Immune-mediated ADRs
  - Rash, serious skin reactions
  - Agranulocytosis, aplastic anemia, thrombocytopenia
  - Hepatitis, hepatic failure
  - Antiepileptic hypersensitivity syndrome (AHS)
- Specific ADRs of each AED

Relative side effects of the antiepileptic drugs

Sedation

Significant	Some
Phenobarbital	Valproate
Carbamazepine	Levetiracetam
Oxcarbazepine	Pregabalin
Phenytoin	Gabapentin
Topiramate	Lamotrigine

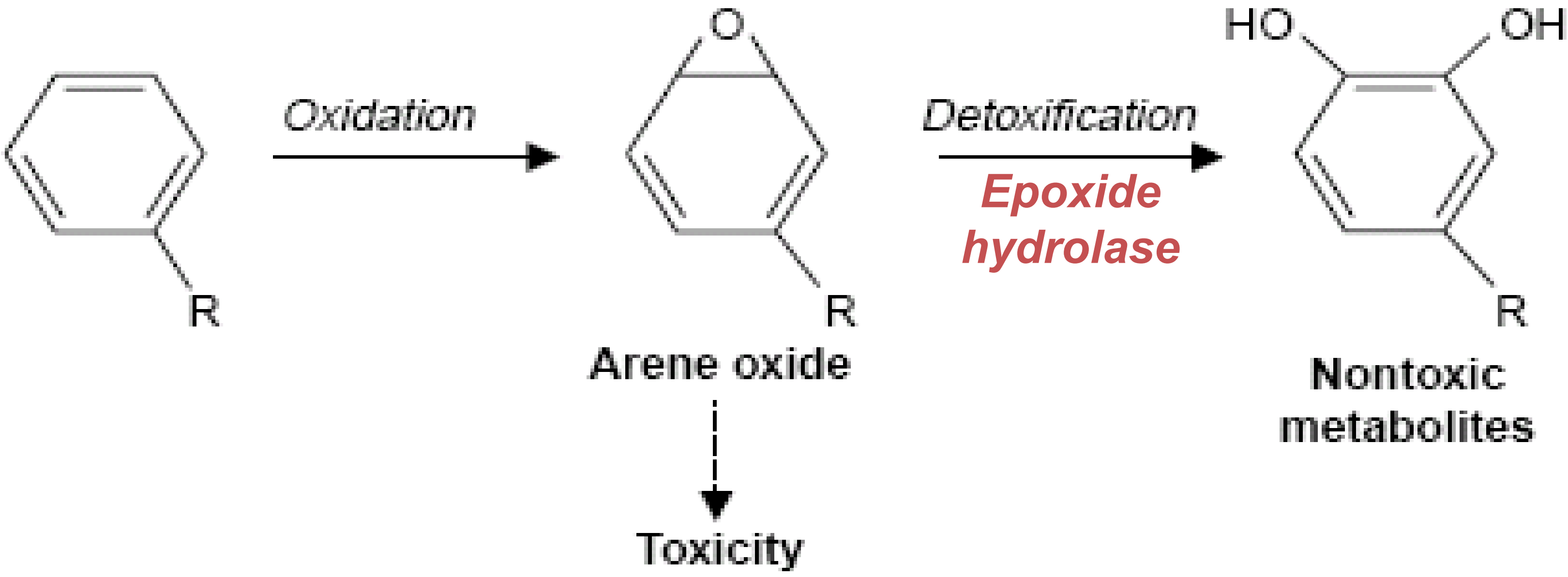
Cognitive side effects

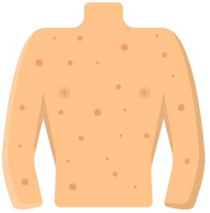
Significant	Some	None or Minimal
Phenobarbital	Carbamazepine	Gabapentin
	Phenytoin	Lacosamide
Topiramate	Valproate	Lamotrigine
	Zonisamide	Levetiracetam
		Oxcarbazepine
		Pregabalin
		Vigabatrin

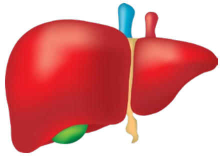
Effects of AEDs on body weight

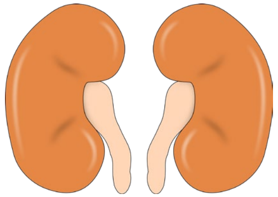
Weight gain	No effect	Weight loss
Carbamazepine Oxcarbazepine Gabapentinoids Valproate	Lamotrigine Lacosamide Levetiracetam Phenytoin Tiagabine	Topiramate Zonizamide Felbamate

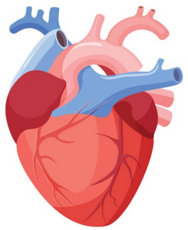
Possible metabolic pathway for production of toxic metabolites of aromatic anticonvulsants → Anticonvulsant hypersensitivity syndrome (AHS)

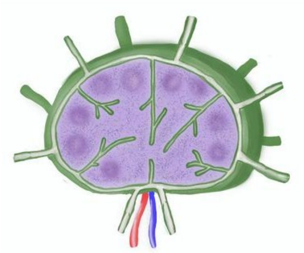


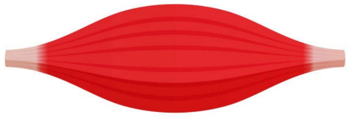








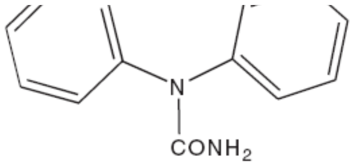




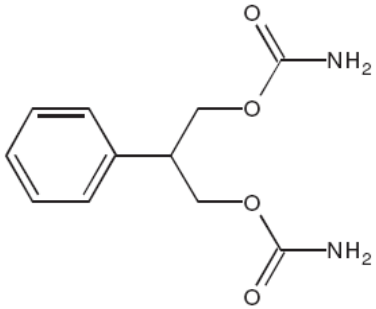
Drug Safety 1999;21:489-501

Aromatic AEDs

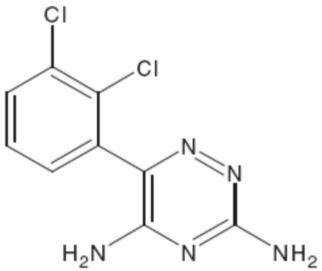
nvulsants



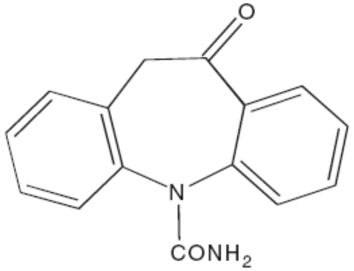
Carbamazepine



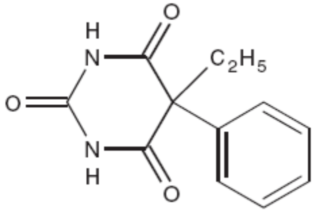
Felbamate



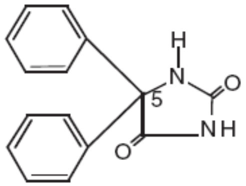
Lamotrigine



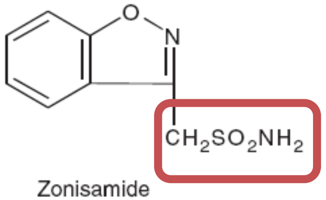
Oxcarbamazepine



Phenobarbital



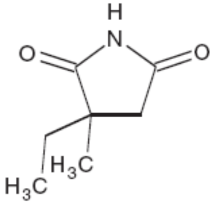
Phenytoin



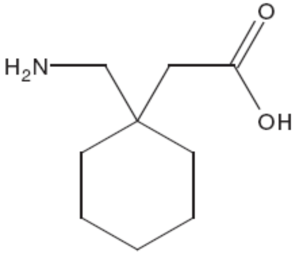
Zonisamide

Nonaromatic AEDs

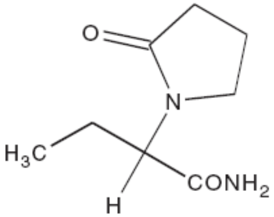
ts



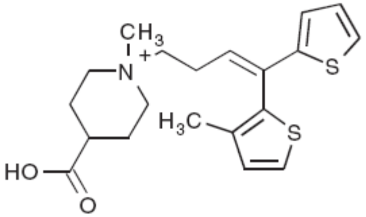
Ethosuximide



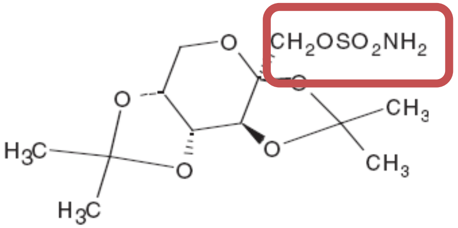
Gabapentin



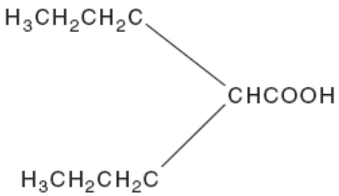
Levetiracetam



Tiagabine



Topiramate



Valproic Acid

Specific adverse effects of commonly used ASMs

- Phenobarbital
  - Reduced IQ
  - Hyperactive child
  - Respiratory depression
- Benzodiazepines (injection)
  - Respiratory depression
- Sodium channel blockers
  - Cardiac rhythm abnormalities
  - Hyponatremia (carbamazepine, oxcarbazepine)
- Phenytoin
  - Gingival hyperplasia, hirsutism, oily skin, acne
- Gabapentinoids
  - Weight gain
  - Peripheral edema
- Valproate
  - Weight gain, alopecia
  - Thrombocytopenia, bleeding risk,
  - Hepatitis, pancreatitis
  - Teratogenicity
- Topiramate
  - Paresthesia, cognitive slowing
  - Kidney stone
  - Glaucoma
  - Weight loss
- Levetiracetam, perampanel
  - Psychosis, behavioral problems

ชื่อยา	ผลข้างเคียงที่พบบ่อย	ผลข้างเคียงสำคัญที่ต้องพึงระวัง	การแพ้ยา
carbamazepine	คลื่นไส้ ซึม เดินเซ เห็นภาพซ้อน	Hyponatremia (SIADH), aplastic anemia, ตับอักเสบ เม็ดเลือดขาวต่ำ	skin rash, Steven Johnson syndrome*
clonazepam	อ่อนเพลีย ง่วง hypotonia พฤติกรรมเปลี่ยนแปลง น้ำลายและเสมหะมาก	กดการหายใจ (ถ้าใช้ยาฉีด)	
gabapentin	ง่วงนอน ซึม เวียนศีรษะ บวม		
lamotrigine	มึนงง เห็นภาพซ้อน เดินเซ		skin rash, Steven Johnson syndrome
levetiracetam	ซึม มึนงง	อารมณ์หงุดหงิด ก้าวร้าว อาการทางจิต	
nitrazepam	ง่วงซึม เสมหะ น้ำลายมาก อ่อนเพลีย hypotonia		
oxcarbazepine	มึนงง ง่วงซึม เดินเซ	hyponatremia	
phenobarbital	เด็ก: ซุกซนไม่อยู่สุข พฤติกรรมเปลี่ยนแปลงก้าวร้าว ผู้ใหญ่: ง่วงซึม อ่อนเพลีย บุคลิกภาพเปลี่ยนแปลง เครียด	serum sickness	skin rash, Steven Johnson syndrome
phenytoin	เวียนศีรษะ เห็นภาพซ้อน ซึม เดินเซ คลื่นไส้ อาเจียน เหงือกบวม หน้าหยาบ hirsutism สิวเพิ่มขึ้น	ตับอักเสบ แคลเซียมต่ำ choreo-athetosis ไข้ และต่อมน้ำเหลือโตทั่วไป เส้นประสาทอักเสบ megaloblastic anemia (folate deficiency) cerebellar degeneration	skin rash, Steven Johnson syndrome

http://thaiepilepsysociety.com/wp-content/uploads/2013/07/CPG_guidelines-for-epilepsy_Edited-at-page-55_Nov-2018.pdf

ชื่อยา	ผลข้างเคียงที่พบบ่อย	ผลข้างเคียงสำคัญที่ต้องพึงระวัง	การแพ้ยา
pregabalin	ง่วงนอน ซึม เวียนศีรษะ		
sodium valproate	มือสั่น คลื่นไส้ อาเจียน ปวดท้อง ผมร่วง น้ำหนักเพิ่ม	ตับอักเสบ ตับอ่อนอักเสบ ภาวะเกล็ดเลือดต่ำ ภาวะ hyperammonemia	
topiramate	มึนงง เดินเซ การพูดผิดปกติ น้ำหนักลด	นิ่วในไต ต้อหิน เหงื่อออกน้อย (oligohidrosis) ความคิดเชื่องช้า ภาวะ hyperammonemia	
vigabatrin	มึนงง ง่วงซึม	ความผิดปกติของลานสายตา	
zonisamide	มึนงง ง่วงซึม เดินเซ เบื่ออาหาร คลื่นไส้	นิ่วในไต ภาวะ agranulocytosis, aplastic anemia	skin rash โดยเฉพาะมีประวัติแพ้ยากลุ่ม Sulfonamide
lacosamide	มึนงง ง่วงซึม ภาพซ้อน เดินเซ	atrioventricular block, palpitation	
perampanel	มึนศีรษะ ง่วงซึม เดินเซ	หงุดหงิด ก้าวร้าว อาการทางจิต มี suicidal ideation	

http://thaiepilepsysociety.com/wp-content/uploads/2013/07/CPG_guidelines-for-epilepsy_Edited-at-page-55_Nov-2018.pdf

Pharmacokinetics and Drugs interactions of Antiepileptics

Pharmacokinetics of conventional antiepileptics

AED	Therapeutic serum conc.	Oral Absorption (%)	Vd (L/kg)	T1/2 (h)	Protein Binding (%)	Elimination	Active Metabolite
Carbamazepine	4–12 ug/mL (CBZ), ,0.2–2.0 ug/mL epoxide)	85	0.8–2	12–17	76	Hepatic (100%): CYP3A4 (major), CYP1A2, CYP2C8	CBZ-10,11-epoxide
Phenobarbital	15-40 ug/mL	70–90	0.5–1.0	36–118	55	Renal (20%): unchanged Hepatic: glucosidase, CYP2C9, CYP2C19, CYP2E1	No
Phenytoin	10–20 ug/mL (total), 1–2 ug/mL (free)	90–100	0.5–1.0	7–42	**90**	Hepatic (98%): CYP2C9 (major), CYP2C19 (minor)	No
Sodium valproate sodium	50-100 (125)[g] ug/mL (total VPA), 5–12.5 ug/mL (free VPA)	100	0.1–0.2	6–17	**90-95** (conc dependent)	Hepatic (95%): beta oxidation, **UGT1A6, UGT1A9, UGT2B7,** CYP2C9, CYP2C19	VPA

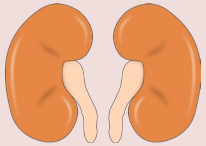
PK disadvantages of conventional AEDs

- Narrow therapeutic index
- High percentage of plasma protein binding (PHT, VPA)
- Mainly metabolized by CYP450
  - PHT PK is non-linear

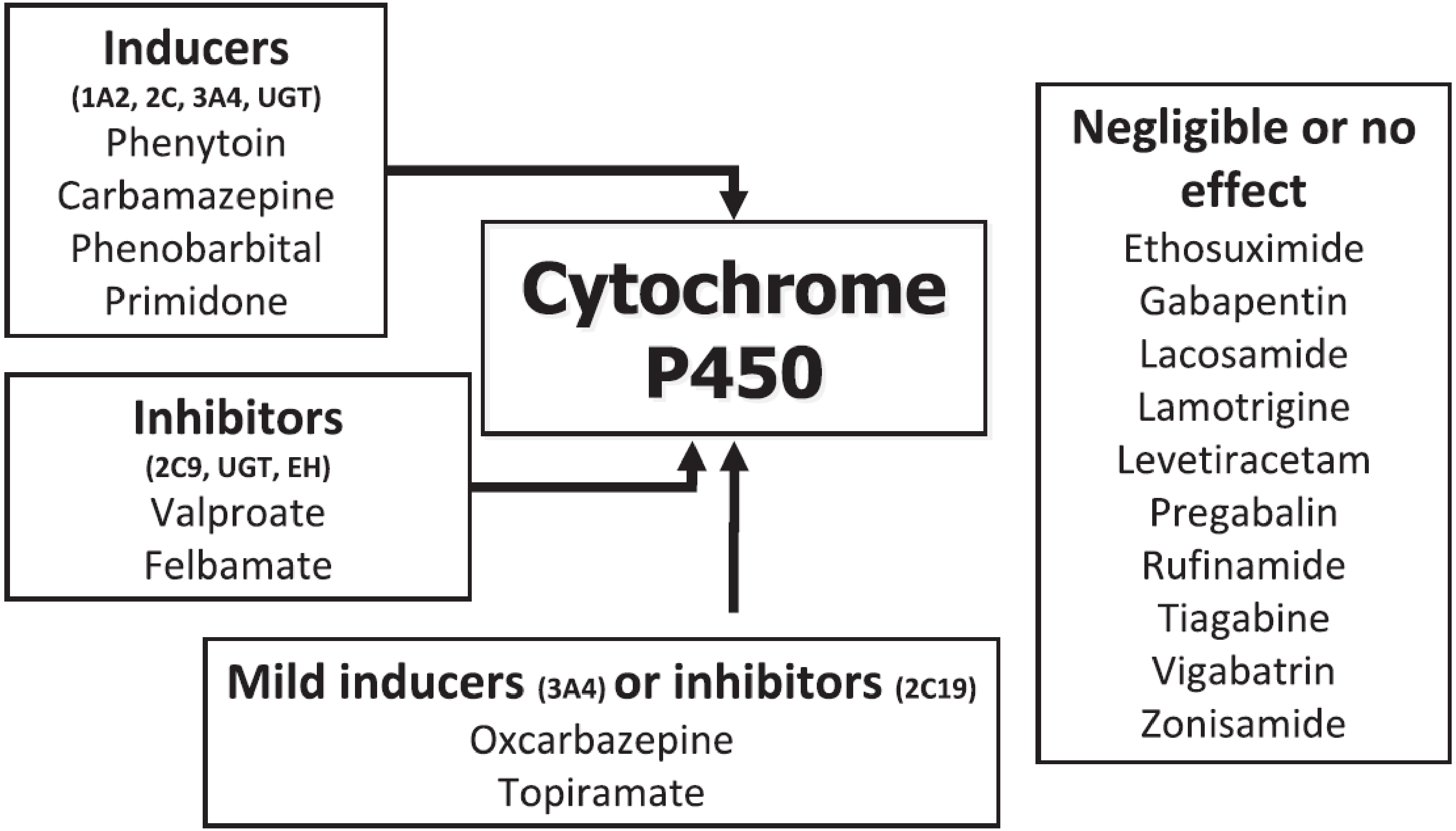
- Induce or inhibit CYP450 activity
  - CBZ, PHT, PB are inducers of CYP450 and UGT
  - VPA is an inhibitor of CYP2C9 and UGT

Major metabolic pathway of AEDs

Metabolism by liver

Predominantly	Partially	Negligible
• Carbamazepine (3A4) • Oxcarbazepine (3A4) • Phenytoin (2C9) • Phenobarbital (2C9) • Valproate • Lamotrigine (UGT)	• Lacosamide (2C19) • Zonisamide (3A4)	• Topiramate • Gabapentinoids • Levetiracetam 

Effects of antiepileptic drugs on the cytochrome P450



Neurol Clin 2010;28:843–852.

AED Inducers

- Induce synthesis of new enzymes
- Slower in onset/offset
- Broad spectrum inducers:
  - Carbamazepine, phenytoin, phenobarbital/primidone
- Concern when using AEDs with enzyme inducing effects

↓ oral contraceptive level

↓ immunosuppressant level (ciclosporin, tacrolimus)

↓ Warfarin level

↓ Folate

↓ Vitamin D & calcium

AED Inhibitors

- Valproate
  - UDP glucuronosyltransferase (UGT)
    - ⇑ plasma concentrations of lamotrigine, lorazepam
  - CYP2C9
    - ⇑ plasma concentrations of phenytoin, phenobarbital
- Topiramate and oxcarbazepine
  - CYP2C19
    - ⇑ plasma concentrations of phenytoin
- Cannabidiol
  - CYP1A2, CYP2B6, CYP2C9, CYP2D6, CYP3A4 and p-glycoprotein
    - ⇑ plasma concentrations of phenytoin, clobazam

Other indication of antiepileptics

- Migraine prevention
  - Valproate, topiramate
- Bipolar disorder
  - Valproate, carbamazepine, lamotrigine
- Neuropathic pain
  - Carbamazepine, oxcabazepine
  - Gabapentin, pregabalin, mirogabalin
- Anxiety disorder
  - Pregabalin